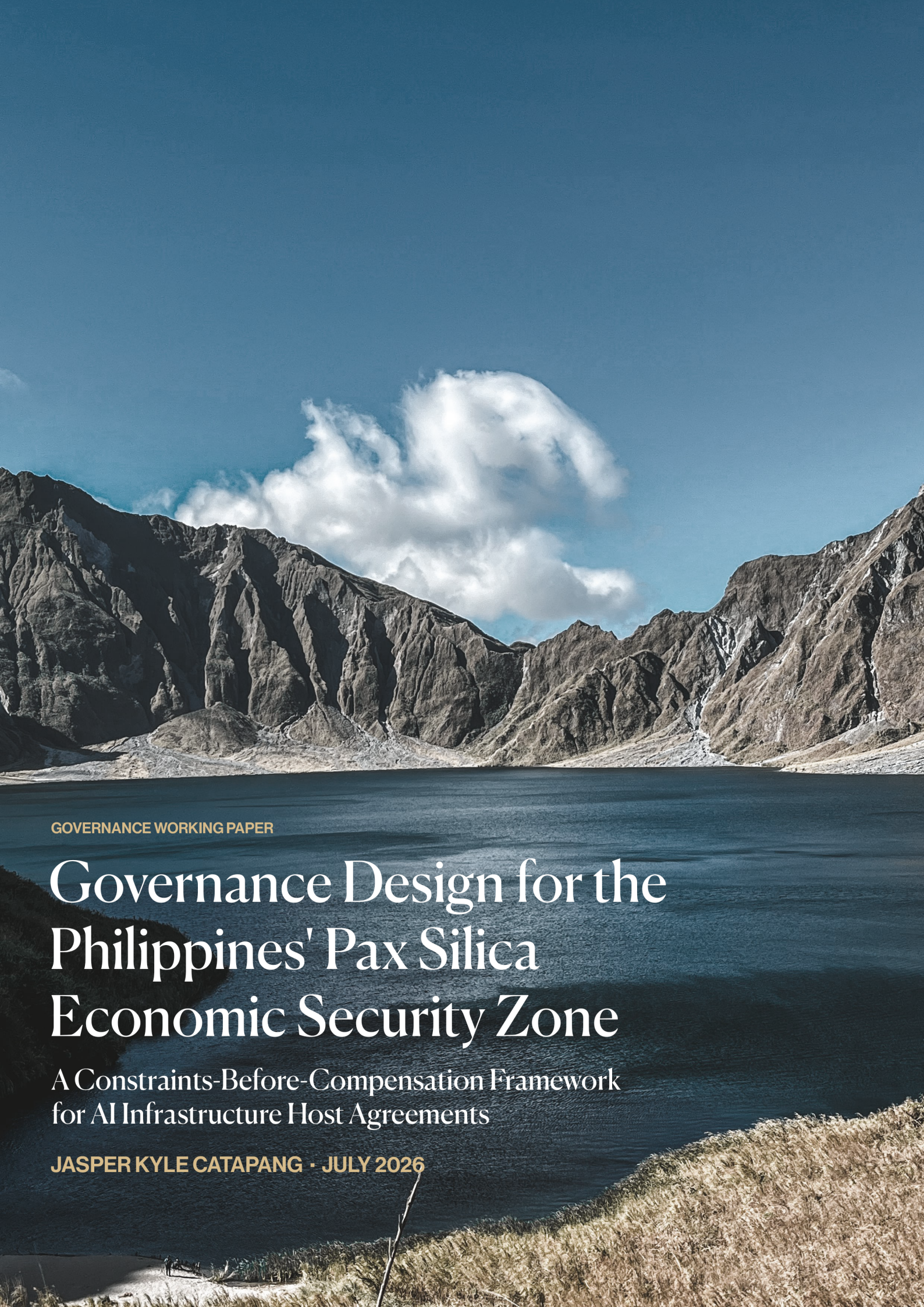




GOVERNANCE WORKING PAPER

Governance Design for the Philippines' Pax Silica Economic Security Zone

A Constraints-Before-Compensation Framework
for AI Infrastructure Host Agreements

JASPER KYLE CATAPANG · JULY 2026

Governance Design for the Philippines' Pax Silica Economic Security Zone

*A Constraints-Before-Compensation Framework for
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Jasper Kyle Catapang
Graduate School of Global Studies,
Tokyo University of Foreign Studies
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Abstract

The November 2026 framework agreement for New Clark City will make the Philippines the first country to host a physical facility of the Pax Silica economic security bloc: a 1,618-hectare Artificial Intelligence (AI) infrastructure zone on a former United States (US) military base in Tarlac Province. These negotiations are not only about whether to proceed, but about what proceeding means.

This paper takes a conditional-proceed position, neither cancellation nor acceleration, occupying the middle ground that no existing government document or coalition has yet articulated rigorously. That vacancy is the paper's reason to exist.

The central analytical device is a three-way cost taxonomy. Compensable costs (relocation, lost income) are answerable with money. Mitigable costs (water stress, grid pressure, pollution) are answerable with engineering, if the agreement requires it. Non-compensable costs (extinguished ancestral domain claims, irreversible aquifer loss, foreclosed ways of life) are answerable by neither. Standard cost-benefit analysis converts the third category into the first by pricing everything; this paper argues the error is analytical, not merely ethical, because it assumes substitutability where none exists.

The corrective is the Constraints-Before-Compensation (CBC) framework: a three-gate evaluation architecture requiring rights preconditions and critical natural capital screening to be settled before any distributional analysis begins. Twelve draft instruments translate the framework into agreement language.

What Manila signs will set the baseline from which every subsequent host negotiates. The Philippines has until November 2026 to write real protections — for its people, land, and laws — into its AI hub deal with the US, not to block the project but to make its benefits real without sacrificing what money can't restore, because whatever it signs becomes the deal every country after it gets offered.

Keywords: AI governance; special economic zones; Pax Silica; indigenous rights; free, prior and informed consent (FPIC); sovereign AI; Philippines; infrastructure hosting agreements

JEL Classification: F21, F53, K32, O25, Q56

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1

Introduction: The First Golden Node as Test Case

In December 2025, the United States assembled an economic security alliance it named Pax Silica, a bloc of partner countries organized around shared supply-chain infrastructure for the semiconductors, sensors, and rare earth-dependent systems at the center of the current technology competition (Shagina, 2026). The organizing logic was direct: China's export controls on critical minerals had demonstrated how concentrated production geography becomes geopolitical leverage in a crisis, and the US intended to redistribute that geography across partner countries whose infrastructure could be integrated with its own. The Philippines formally joined in April 2026. Within weeks, the Bases Conversion and Development Authority announced that New Clark City in Tarlac, 1,618 hectares of former US military land administered under the Bases Conversion and Development Act of 1992, Republic Act (RA) No. 7227, would serve as the initiative's first physical AI hub: a 4,000-acre economic security zone built specifically to anchor Pax Silica's hardware and processing operations (U.S. Embassy in the Philippines, 2026; Bases Conversion and Development Authority, 2026). Ground-breaking is expected before 2028 (Ayeng, 2026). A framework agreement governing the zone's legal structure, investment terms, and the relationship between Philippine law and the zone's operations is expected by November 2026. Nothing in the Pax Silica bloc is further along than this. New Clark City is the first planned physical facility of the initiative anywhere on earth.

That timing defines the moment this paper addresses. Whatever the Philippines writes into the November 2026 framework agreement, or leaves unwritten under the time pressure of negotiation, becomes the founding document of an arrangement that will govern the site for decades. It also becomes the observable template from which every subsequent Pax Silica host country will negotiate its own participation. First movers in novel institutional arrangements bear disproportionate responsibility for the terms that follow, because the provisions they accept, under deadline pressure and information asymmetry, become the provisions later parties are offered as a baseline. Philippine leverage over those terms is highest now, before the agreement is signed and before the first phase of construction commits both parties to a path. That leverage diminishes with each milestone that passes without explicit protections in place.

Official communications from both governments have emphasized the project's scale and expected economic returns. The investment interest reported has been tracked inconsistently across sources and dates: roughly 20 firms were noted in May 2026 (Gonzales, 2026), a figure that appeared as approximately 30 by late July (Romero, 2026) and as approximately 50 in a separate report from mid-June (Domingo, 2026), while BCDA projected total investment of between \$40 billion and \$70 billion across the same period (Sinaking, 2026). Employment projections have moved on a similar spread, from figures above 100,000 in April 2026 (Ayeng, 2026) to figures approaching 190,000 by late July (Romero, 2026). No published document sets out the methodology behind any of these numbers, the assumptions about direct versus induced employment, or the period over which the jobs are expected to materialize. The inconsistency across every category (firms, investment, livelihoods) is not simply a product of evolving information in a fast-moving negotiation. It reflects an aggregation process whose bookkeeping, by the public record's own evidence, has not been rigorously standardized. A ledger that cannot agree with itself on headline figures warrants scrutiny before it is used to weigh benefits against costs.

The costs in question are not speculative. New Clark City is administered under RA 7227, which created BCDA and established the Clark Special Economic Zone; RA 7916 § 46 expressly excludes BCDA and RA 7227 areas from the scope of the Philippine Economic Zone Authority (PEZA) law. The land's history is long. Clark was a major US military installation whose closure left a legacy of environmental contamination that remediation programs have addressed only partially (Pure Earth, n.d.). The communities nearest to New Clark City include Aeta indigenous families, specifically the Aeta Hungey tribe, approximately 500 families in Sapang Kawayan, who have been contesting displacement pressures that predate the Pax Silica announcement, as well as farming households in Capas facing related pressures (Beltran, 2025). Ancestral domain claims are pending. Reports of harassment of Capas residents surfaced in July 2026 (SunStar Pampanga, 2026). These conditions do not appear in official project communications from either government, a silence this paper does not interpret as evidence of intention but does treat as evidence of a gap in the governance design.

The project's potential benefits carry genuine weight and must be taken seriously as such before any accounting discipline is applied to them. Investment at any point in the range described would represent one of the larger

industrial commitments in Philippine history. The geopolitical argument is coherent: shared supply-chain infrastructure insulates partner economies from the leverage that concentrated production geography can generate, and it serves Philippine interests as well as American ones. The Philippine government's participation reflects a real strategic calculation: the country has a longstanding interest in attracting capital-intensive, technology-adjacent investment of exactly this kind, and Pax Silica offers the additional possibility of technology and skills transfer if the agreement's provisions require it. If the employment projections associated with the project materialize anywhere in the range cited, the effect on household income and regional economic structure in Central Luzon would be significant in ways that critics of the project have not always fully engaged. The initiative's architects have described it as partnership, and the distinction between partnership and extraction is not settled by vocabulary. It is settled by provisions.

This paper argues that a rigorous, publicly available account of what a conditionally favorable agreement would look like does not currently exist, and that its absence has a specific cost. The Philippines has until November 2026 to write real protections for its people, land, and laws into the agreement, not to block the project but to make its benefits real without sacrificing what money cannot restore, because whatever the Philippines signs becomes the deal every country after it will be offered. That argument, made with the specificity the moment requires, has not been made. The Philippine government has advanced the project. Critics have called for suspension or cancellation. What neither position has produced is a detailed account of the provisions a legitimate framework agreement must contain: what governance architecture, what legal safeguards, what procedural requirements separate a deal that works for the Philippines from one that primarily works for others. This paper occupies that vacant position, not because equidistance is a virtue in itself but because the conditional-proceed analysis is the correct one. Many of the project's benefits are achievable with the right instruments in place. Most of its risks are manageable if addressed before rather than after signing. A small category of potential losses (to ancestral domain, to ecological function, to ways of life that cash transfers cannot reconstruct) are neither reversible nor compensable once they occur, and can only be avoided by writing preconditions into the agreement before ground breaks.

The skeptical response to this initiative has a documented historical basis. The September 1991 Senate vote — 12 to 11, with Senate President Jovito Salonga casting the deciding vote — declined to renew the US military bases treaty (Simbulan, 2018). Three decades later, Pure Earth documents ongoing remediation of contamination at the Clark and Subic Bay sites that arrangement left behind (Pure Earth, n.d.). The Aeta Hungey families of Sapang Kawayan carry unresolved ancestral domain applications predating the Pax Silica designation; the May 2026 immunity episode raised direct questions about the terms on which foreign personnel would operate on Philippine soil (Beltran, 2025; Chi, 2026, May 19). These are not abstract concerns, and they have precedent.

The structural situation is also documented. The Philippines acceded to Pax Silica in April 2026 as a sovereign decision. The Mutual Defense Treaty, the Visiting Forces Agreement, and the Enhanced Defense Cooperation Agreement — upheld 10 to 4 by the Supreme Court — are the legal architecture within which the current negotiation sits. BCDA's investment projection reaches \$70 billion (Sinaking, 2026); 50 firms have signaled interest (Domingo, 2026); ground-breaking is projected before 2028 (Ayeng, 2026). The framework agreement is proceeding.

The conditional-proceed position follows from both sets of facts. Where the historical record supports skepticism about the terms of the arrangement, it also supports the argument that the terms must be specified, not avoided. Where the structural situation forecloses cancellation as a practical outcome, it also concentrates the available leverage in the period before signing. The analysis that follows takes both of those implications seriously.

The case for that conclusion requires working through the zone's legal and political genealogy, through the governance provisions that official communications leave unresolved, through an examination of what the standard evaluative framework systematically omits, and into a set of specific instruments that could be written into the agreement before November 2026 closes. The goal throughout is a document that earns the engagement of a BCDA official, a State Department staffer, and a Kalikasan organizer alike, and that earns it by taking all three seriously.

2

Genealogy: How the Zone Got Here

In the years leading up to December 2025, China progressively tightened export controls on rare earth elements and critical minerals that are foundational to semiconductor fabrication, battery production, and the hardware infrastructure of large-scale artificial intelligence. Building on the legal architecture of its 2020 Export Control Law, Beijing introduced licensing requirements for gallium and germanium in July 2023, extended them to graphite in October 2023 and to rare-earth processing technologies that December, added antimony and superhard materials in August 2024, and in December 2024 moved from licensing to prohibition for shipments of gallium, germanium, and antimony to the United States. Controls on seven rare earth elements followed in April 2025. The decisive measure came on October 9, 2025, two months before Pax Silica was announced, when the controls were extended beyond physical goods to cover related technologies and know-how, and to foreign-made products incorporating Chinese inputs (Global Trade Alert, 2025a, 2025b; ORF America, 2025). The effect on US industrial and defense planning was acute: a technology-supply architecture built on the assumption of Chinese material flows became legible as a strategic vulnerability. Pax Silica was the American response to that vulnerability: a formal grouping of allied and partner nations organized around the goal of establishing alternative supply chains for the materials undergirding the AI technology stack. As Shagina (2026) traces in her analysis for the International Institute for Strategic Studies, the initiative represents a distinct phase in US critical-minerals diplomacy, a movement from bilateral purchase arrangements toward what she characterizes as allied supply-chain territorialization: partner countries integrated into a geographically distributed, US-anchored network of production, processing, and infrastructure that reduces exposure to Chinese chokepoints.

The Philippines formally acceded to Pax Silica in April 2026. Under the terms announced jointly by the U.S. Embassy in Manila and the Bases Conversion and Development Authority, 1,618 hectares within New Clark City in Tarlac province would be designated as the initiative's first physical facility anywhere in the world (U.S. Embassy in the Philippines, 2026; BCDA, 2026). Administration of the zone would fall to the BCDA, the authority established by the Bases Conversion and Development Act of 1992 (Republic Act No. 7227), which remains the controlling statute for New Clark City. Ground-breaking was projected to occur before 2028 (Ayeng, 2026). The BCDA, which had spent more than three decades converting former American military installations into civilian economic platforms, would now steward the Philippines' formal entry into the Pax Silica architecture.

The case for that entry is not trivial, and it deserves a fair statement before any part of it is questioned. BCDA's own projections place the zone's cumulative investment potential at between \$40 billion and \$70 billion, with employment estimates ranging from 100,000 to more than 190,000 jobs across sources and dates (Sinaking, 2026; Romero, 2026). For a province that carries the economic legacies of regional crises and agricultural decline, those figures represent a structural transformation rather than a marginal addition. The US and Philippine governments have framed participation in Pax Silica as a means of positioning the Philippines within the higher-value segments of the global AI supply chain, rather than remaining a supplier of raw or minimally processed critical materials. A successfully governed zone could anchor technology-transfer obligations, skills pipelines, and fiscal revenue flows that would outlast any single administration on either side of the agreement. The project's supporters are making a coherent argument, and the analysis that follows proceeds on the assumption that the outcomes they describe are ones they genuinely intend to deliver.

What the official framing does not address is that the 1,618 hectares carries a history, and that history is not finished. Clark has been American in one form or another for most of the past century. Established as a military installation during the American colonial period, Clark Air Base grew over the following decades into a major US Air Force facility and the anchor of the American military presence in the archipelago. The 1991 eruption of Mount Pinatubo deposited lahar across the surrounding plain and severely damaged the base's infrastructure; the following year, after the Senate vote of September 16, 1991 declined to ratify a new basing treaty (Simbulan, 2018), the United States withdrew and the installation was transferred to the Philippine government under the mechanisms of RA 7227. The legal architecture of the site was revised again in 2007, when RA 9400 separated the former air base into the Clark Freeport Zone and extended RA 7916 incentive provisions to designated areas within it, leaving the wider Clark Special Economic Zone, and with it New Clark City, under BCDA administration. The formal exit was clear enough. The environmental accounting was not.

Pure Earth, the international environmental health organization, has documented ongoing remediation work at both Clark and Subic Bay, where hazardous substances accumulated across decades of military operations, including heavy metals, petroleum hydrocarbons, and other industrial contaminants, continue to pose environmental and public health risks (Pure Earth, n.d.). The remediation is active, not concluded; the affected land has not been certified clean. This matters not as an argument against development but as a statement of fact about the site's condition: the 1,618-hectare parcel designated for Pax Silica sits within the broader Clark footprint where this contamination legacy remains an open file.

Under RA 7227 and Proclamation No. 163 of 1993, which created the Clark Special Economic Zone and placed it under BCDA administration, the former base was formally repositioned as a platform for civilian economic development. New Clark City, the BCDA's most ambitious project within that frame, was conceived as a planned urban center in the municipalities of Capas and Bamban, designed to relieve metropolitan congestion and attract both domestic and foreign investment. By the time Pax Silica's designation arrived in April 2026, New Clark City was already a project with its own contested developmental history. Roads, athletic facilities, and government buildings had already been constructed or were underway. The AI hub is therefore not being placed on bare or undeveloped ground. It is being placed on a development project that was itself layered onto a military installation whose environmental remediation is incomplete and whose conversion from military to civilian use generated conflicts that had not been resolved when the next layer of ambition arrived.

Those conflicts were live and documented before any agreement bearing the Pax Silica name was signed. Reporting by Al Jazeera in December 2025, four months before the Philippines' formal accession, documented an active dispute involving the Sapang Kawayan community, home to members of the Aeta Hungey tribe, one of the indigenous peoples whose ancestral territories historically encompassed the Clark area (Beltran, 2025). Approximately 500 families faced displacement as New Clark City construction expanded across land where their ancestral domain applications remained unresolved. Those applications were filed under the Indigenous Peoples' Rights Act of 1997 (Republic Act No. 8371), which establishes free, prior, and informed consent as a domestic legal condition for any activity affecting indigenous lands. Under that Act, the absence of a Certificate of Ancestral Domain Title does not extinguish customary land rights; it records only that the state has not yet completed the documentation of rights that the statute already recognizes. The same reporting identified a second, distinct source of pressure: a proposed \$515 million stadium project, also planned within the New Clark City footprint, would accelerate and compound the land displacement these communities already faced (Beltran, 2025). The AI hub, when designated, became a third source of pressure on the same population.

Indigenous communities are not the only affected group. A July 2026 report by SunStar Pampanga documented that residents of Capas had organized to protest what they described as harassment connected to the ongoing development, and that the 790th Air Base Group's Joint Training Complex at Crow Valley had generated a further, distinct set of land-use pressures on farming communities in the area (SunStar Pampanga, 2026). Crow Valley, the US military training range that has operated in Tarlac through and beyond the base era, represents a line of pressure separate from the Aeta ancestral domain dispute and separate from the AI hub designation itself. The Capas farmer situation, the Aeta displacement conflict, the unremediated contamination, and the Crow Valley training-complex pressure are not a single problem; they are several problems that share the same communities and the same terrain. Cumulatively, they constitute an impact picture substantially larger and more complex than the 1,618-hectare polygon that the official designation documents describe.

Three layers of history now share the same ground: the unfinished remediation of a military installation, the contested conversion of that installation into a planned city, and the designation of part of that planned city as the world's first Pax Silica node. Each layer arrived before the previous one was resolved. The Aeta ancestral domain claims predate New Clark City's construction; the community displacement conflicts were already live before Pax Silica was announced; the contamination from the base era is under active remediation as the zone is being planned. The zone is not being built on empty land; it is being built on an unfinished argument about the last arrangement. The ledger the Philippine government presents as justification for proceeding makes this history invisible, not because it is dishonest, but because the accounting method cannot see what it cannot price.

The significance of this genealogy is not that it forecloses development. The conversion of former military land into civilian economic use has generated genuine gains at multiple sites across the Philippines since 1992, and there is no principled objection to adding a technology-investment mandate to a zone that already exists. The significance is that the history is unresolved, and the framework agreement now being negotiated will either engage that reality or will not. An agreement designed as though the underlying land disputes have already been settled, as though the contamination legacy is already closed, as though the communities contesting earlier layers of the project have been heard, would be an agreement about an imagined site. The site that actually exists comes with prior claims, incomplete remediation, and communities that have been in open conflict with successive development projects for years. Any governance framework adequate to those conditions has to begin by acknowledging them.

3

The Governance Problem: Whose Law, Applied by Whom, to Whom

The episode that most clearly exposed the framework agreement's unresolved interior arrived on May 19, 2026, when Manila declined a reported United States request for diplomatic immunity covering personnel slated to operate at the planned Clark AI hub (Chi, 2026, May 19). The Philippine government's refusal was publicly stated, and the reporting that followed characterized the exchange as a significant jurisdictional confrontation. Six days later, Jacob Helberg, the Under Secretary of State for Economic Growth, Energy, and the Environment, pushed back in a post on X, contesting the framing of the request and arguing that the nature of what had been sought had been misrepresented; the account, he indicated, fundamentally misunderstood the partnership logic underlying Pax Silica (Helberg, 2026, May 25). A follow-up report noted that the US position was that the immunity request had been taken out of context (Chi, 2026, May 26).

Helberg's objection deserves a fair hearing before it is answered. The diplomatic record that would permit a definitive adjudication of precisely what was asked is not in the public domain, and extending interpretive charity to both sides requires declining to treat a contested account as established fact. What the episode demonstrates, even under the most favorable reading of his position, is that the question of which sovereign's law governs, applied by whose institutions and enforceable against whom, was sufficiently unresolved in May 2026 that a public dispute about it could arise at all. The episode is diagnostic not because it confirms the harshest interpretation but because it reveals a design space that remains, as of this writing, formally unaddressed.

The honest analysis also requires acknowledging what the critics' version typically elides. Philippine and American forces already operate under a jurisdictional regime that allocates criminal jurisdiction primarily to the United States for offences committed by its military personnel on Philippine soil. That arrangement traces to the foundational Mutual Defense Treaty of August 30, 1951, was formalized in the Visiting Forces Agreement of February 10, 1998, which entered into force on June 1, 1999, and was extended and operationally deepened by the Enhanced Defense Cooperation Agreement of April 28, 2014. EDCA was constitutionally challenged almost immediately and upheld by the Philippine Supreme Court on January 12, 2016 in *Saguisag v. Ochoa*, G.R. Nos. 212426 and 212444, which held EDCA to be an executive agreement implementing the Mutual Defense Treaty and the VFA and therefore not requiring separate Senate concurrence. These instruments represent a durable Philippines-US jurisdictional arrangement that has survived both legal challenge and shifts in the executive. A request to extend analogous protections to civilian and commercial operators at Pax Silica is therefore not a categorical rupture with Philippine legal practice; it is the transposition of an existing military-domain pattern into the civilian-commercial domain. Treating it as unprecedented overreach misreads the jurisdictional history. Acknowledging the transposition is precisely what makes the diagnosis useful rather than rhetorical: the question is not whether the Philippines has accommodated US jurisdictional preferences before, because it has and substantially, but whether the same accommodation, moved from its security context into an explicitly commercial one, carries equivalent justification without the procedural architecture that would render it publicly defensible.

That architecture, in its specifics, is what the May 2026 episode cannot contain by itself. The immunity question was the most visible surface of a considerably wider design space.

A survey of publicly available documentation reveals that the terms on which Pax Silica will operate at New Clark City are, across most consequential dimensions, not yet publicly specified. The legal regime governing land acquisition is not settled in any published document: who holds title, under what conditions conversion from current BCDA-administered status proceeds, and what security of tenure communities in the affected area retain all turn in part on the fate of H.B. 9257 of the 20th Congress, which would amend RA 7227 to declare portions of BCDA-managed zones including New Clark City alienable and disposable. The measure was reported out as Committee Report No. 300 of the 20th Congress, which sets out the proposed Section 15-D in full (House of Representatives, 2026). The House approved it on second reading on May 26, 2026 and on third and final reading on June 2, 2026, by a vote of 271 in favor, nine against, and two abstentions (Politiko, 2026; Manila Bulletin, 2026). As of July 26, 2026, no record of Senate concurrence or enactment has been located in the Congress of the Philippines legislative information system; the measure's final disposition, and its timing relative to the framework agreement, remain open. In parallel, H.B. 4663, the proposed New Clark City Freeport Area Act filed in the 20th Congress by Representatives Jaime Eduardo Marc D. Cojuangco and Noel N. Rivera, would establish a distinct New Clark City Development Authority and apply RA 7916 incentive provisions within a

defined freeport perimeter. It has been filed and its text is publicly available, but no record of committee action, plenary consideration, or passage could be located; on the available evidence it remains at an early stage, and its prospects are unknown. That two pieces of legislation directly reshaping the land and authority framework of the zone are moving concurrently with the framework agreement negotiations, without public acknowledgment of the interaction, is itself a governance observation.

Labor standards applicable to what BCDA projects will become a workforce of between 100,000 and 190,000 people, a figure whose wide range across published sources is addressed separately in Part II, are unspecified in the public record. Environmental review requirements, their scope and independence, and the identity of the body that commissions and publishes them remain unaddressed. Dispute resolution procedures, including whether communities not only investors will have enforceable access to binding remediation mechanisms, carry no published specification. Data governance, covering what is processed at the hub, by whom, subject to which national law, and under whose security protocols, has received no public articulation. The relationship between hub operations and Philippine security integration, and specifically how that integration is managed given Pax Silica's explicit origin as a response to geopolitical contest (Shagina, 2026), is unspecified; so are water allocation rules for a site situated above the Pampanga aquifer system, energy supply arrangements and the allocation of grid-buildout costs, revenue-sharing formulas between the zone authority and national and local government, and the monitoring and transparency obligations of zone operators. A structured accounting of these settled and unsettled provisions, organized by category and noting the applicable legal instruments where they exist, appears in Annex A.

The pattern these gaps share is not necessarily evidence of bad faith on either side. Framework agreements characteristically defer technical specification to subsequent negotiating rounds, and this is accepted practice. What distinguishes the present situation is the combination of timeline compression and the existence, in domestic Philippine law, of requirements that do not wait on subsequent rounds to be operative.

The Philippines has navigated versions of the jurisdictional design problem before and across different registers, and that experience provides useful reference rather than simple precedent. The US bases regime's evolution from the Mutual Defense Treaty through the VFA to EDCA established the template within the security domain: bilateral agreements allocating jurisdiction, constructing access rights, and specifying enforcement mechanisms, with Philippine courts retaining formal authority but practical primacy sitting elsewhere in defined circumstances. The EDCA upheld in 2016 and the VFA in force since 1999 are the closest structural precedents for how executive-agreement frameworks have handled Philippine-US jurisdictional allocation. Special economic zones under RA 7916, the legal basis for the Philippine Economic Zone Authority's jurisdiction, offer a second comparative register, covering how the Philippines has previously constructed investor-protection and governance frameworks within bounded geographic areas. The United Nations Conference on Trade and Development (UNCTAD) World Investment Report 2019, which surveyed approximately 5,400 zones across 147 economies, and the World Bank assessments by Zeng (2015, 2021) both document the recurring pattern in which investor protections in special zones are substantially more specific and enforceable than the protections available to host communities, a pattern that has contributed to the dispossession dynamics Levien (2012) analyzed in the Indian SEZ context. These international comparators are cited as context, not as governing law: PEZA's RA 7916 does not govern New Clark City, because Section 46 of that act expressly excludes zones established under RA 7227 and areas administered by BCDA from PEZA's authority (Special Economic Zone Act of 1995, § 46). The controlling statute is RA 7227, the Bases Conversion and Development Act of 1992, and it already contains provisions directly relevant to the governance questions the framework agreement must resolve.

Section 15 of RA 7227 provides that zone policies "shall be determined upon consultation with the inhabitants of the local government units directly affected," and that zone creation is "subject to the concurrence by resolution of the local government units directly affected" (Bases Conversion and Development Act of 1992, § 15). H.B. 4663, in its own Section 3, carries an identical LGU-concurrence requirement for the proposed New Clark City Freeport Area. These are statutory conditions, not aspirational guidelines. The municipalities of Capas and Bamban sit within the zone's direct footprint. Whether those consultation and concurrence obligations have

been discharged, and how any framework agreement will document that discharge, is a domestic legal question that precedes any normative argument about the desirability of more inclusive governance.

The constitutional superstructure reinforces the same reading. Article XII, Section 10 of the 1987 Constitution provides that “the State shall regulate and exercise authority over foreign investments within its national jurisdiction” and that “in the grant of rights, privileges, and concessions covering the national economy and patrimony, the State shall give preference to qualified Filipinos.” The Philippine Supreme Court held in *Manila Prince Hotel v. GSIS* that Section 10, paragraph 2 is self-executing, requiring no implementing legislation; it is a direct constitutional constraint on the terms on which foreign concessions may be granted. Article XII, Section 2 reserves all lands of the public domain, waters, and natural resources under full State control and supervision. The Free, Prior and Informed Consent (FPIC) question carries its own domestic architecture: the Philippines has not ratified International Labour Organization (ILO) Convention No. 169, the Indigenous and Tribal Peoples Convention, which sets the international benchmark for free, prior, and informed consent; Philippine FPIC obligations derive instead from the Indigenous Peoples’ Rights Act of 1997, RA 8371, the framework under which the Aeta communities’ claims in the zone’s footprint must be assessed (Beltran, 2025). ILO C169 is the standard Manila has declined to bind itself to, a position that is itself a finding, since it signals a deliberate choice about the level of international obligation the Philippines will accept in this domain. The United Nations Declaration on the Rights of Indigenous Peoples, adopted in September 2007, is legally non-binding as a General Assembly resolution; the United States was one of four states that voted against its adoption in 2007, a position all four later reversed (G.A. Res. 61/295, 2007).

The jurisdictional questions that surfaced in May 2026 are expressions of a structural feature of the arrangement rather than administrative oversights. The framework agreement, as it exists in public documentation, specifies the zone’s aspiration without specifying its operating architecture. The aspiration, a world-class AI supply chain embedded in a rules-based economic security bloc, is not the contested element. The question is whose rules, applied by whose institutions, with recourse available to whom. Sovereignty is decided in procedural details, not declarations; a framework agreement that defers those details to later rounds while conversion and site preparation proceed has answered the question by deferral, and that answer has consequences the deferral cannot undo.

The asymmetry that results deserves to be stated plainly. Investment protection under the zone framework, as supplemented by prospective freeport legislation and by bilateral economic agreements, provides investors with defined rights against expropriation, enforceable dispute resolution, and protections that travel with the capital into whatever legal form the zone eventually takes. The parallel protections available to communities are: the consultation and LGU-concurrence rights under RA 7227 § 15, the FPIC rights under IPRA, and the constitutional standing under Article XII. These exist in domestic law but lack, in the current framework, any mechanism for enforcement symmetrical with what investors will hold. This is not a critique of investment protection as such; it is a description of how the legal architecture currently operates within a single geographic zone. Asymmetry does not cease to be asymmetry because it is described in the language of partnership, and the description does not alter the design. Legal certainty for investors is a legitimate need; the question the May 2026 episode leaves open is whether blanket immunity is the only mechanism capable of providing it, or whether a properly designed grievance architecture achieves the same result within Philippine jurisdiction.

4

The Risk Ledger the Ledger Ignores

Three categories of risk are absent from official communications about the Pax Silica Clark hub: from the BCDA press release, the US Embassy announcement, and the public investment-attraction materials that accompanied escalating interest from approximately twenty firms in May 2026 to fifty in June 2026 (Gonzales, 2026; Domingo, 2026; Romero, 2026; BCDA, 2026; U.S. Embassy in the Philippines, 2026). Their absence does not constitute an argument for halting the project. It is an observation that the decision matrix presented publicly omits factors that any thorough risk assessment would include, and that omitting them does not make them less real.

The New Clark City site sits within the lahar-affected region of Mount Pinatubo, whose 1991 eruption reshaped the surface hydrology and land form of Pampanga and Zambales provinces across hundreds of square kilometers. The Philippine Institute of Volcanology and Seismology maintains published hazard maps for both provinces, the Pinatubo Volcano Hazard Map of Pampanga and the Pinatubo Volcano Hazard Map of Zambales, which delineate exposure zones for lahar flows, pyroclastic density currents, and ashfall in scenarios involving renewed volcanic activity (PHIVOLCS, n.d.-a; PHIVOLCS, n.d.-b). No publicly available zone document, environmental scoping material, or framework agreement text reviewed for this paper addresses those maps or their implications for site selection, infrastructure design, or emergency management planning at the proposed scale. This is not a new geotechnical claim; the hazard is mapped, published, and official. The finding is the absence of engagement: a major infrastructure investment sited in a known volcanic-hazard region, with no public accounting of that exposure in the documentation governing the zone.

The second risk category concerns geopolitical positioning. The integration of the Clark hub into the Pax Silica supply-chain security framework, a bloc whose formation was a direct response to China's rare earth export controls (Shagina, 2026), is the feature that distinguishes this AI zone from a generic special economic zone and also the feature that creates a strategic exposure official communications leave unaddressed. Infrastructure explicitly positioned within a named US-led economic security architecture is, by that positioning, embedded in a strategic contest whose direction communities in Capas and Bamban cannot determine and were not consulted about. The sober version of this observation is probabilistic, not alarmist: in a scenario of significant regional military tension, civilian infrastructure that forms part of a designated US-aligned supply-chain security initiative occupies a different risk profile than infrastructure without that designation. The communities who live in proximity to that infrastructure bear the consequences of the alignment without having been party to the alignment decision. That asymmetry belongs in any complete risk disclosure, and its absence from public documentation is a governance finding regardless of the probability one assigns to the underlying scenario.

The third risk category operates entirely within the logic of financial prudence and is the one most likely to concentrate the attention of investors and both governments, precisely because it does not require accepting any ethical premise to take seriously. The AI infrastructure buildout that Pax Silica represents is occurring at a moment of extraordinary capital intensity. BCDA's own estimate places prospective investment at \$40 to \$70 billion (Sinaking, 2026), a projection that reflects assumptions about the durability of current data-center demand growth, the continuity of US industrial policy toward allied supply-chain investment, and private-sector appetite for long-horizon infrastructure commitments in jurisdictions linked to a particular geopolitical configuration. Each of those assumptions is contestable on its own terms and more contestable in combination.

AI capital expenditure cycles have historically been subject to rapid reassessment when benchmark expectations shift, when competing infrastructure configurations emerge in other jurisdictions, or when the consensus technology architecture changes in ways that alter the physical requirements of compute infrastructure. The US industrial policy that Pax Silica instantiates, meaning the specific blend of export controls, allied investment facilitation, and bilateral security-economic integration, reflects the political configuration of a particular moment; its successors will be shaped by people and circumstances not yet determined. If either variable shifts materially before the zone reaches build-out, the Philippines will hold converted agricultural land, infrastructure at various stages of completion, and the fiscal and social costs of a displacement and conversion process that is not reversible on any meaningful timescale. Brownfield infrastructure in a converted zone is not a recoverable asset if the demand that justified the conversion has moved elsewhere or ceased to materialize.

This stranded-asset scenario is not a reason to oppose the zone; it is a reason to structure the zone's development so that no party is maximally exposed to the downside before the upside has been demonstrated. Phasing de-

velopment, making later construction stages legally conditional on published performance benchmarks achieved in earlier ones and independently certified, is not a concession to critics of the project. Phasing is insurance for investors. If the AI capex cycle turns, nobody wants to own a stranded zone, and a phased structure limits exposure for host and investor alike while preserving the full upside if the projections hold. The financial case for phasing is independent of, and prior to, any social or environmental rationale; it stands on grounds that speak directly to the self-interest of both governments and every investor currently expressing interest.

The official ledger presents job projections, investment-interest tallies, and supply-chain positioning. It ignores costs it cannot price and risks it prefers not to see, and those two blind spots compound rather than cancel. A framework agreement silent on both will not produce the durable, enforceable arrangement its negotiators say they want. The costs the official ledger cannot see are the subject of Part II. The instrument that converts the stranded-asset risk into design discipline is the phased development gate, not as a concession to critics but as insurance that both parties have agreed to underwrite.

5

The Evaluation Problem: What the Ledger Cannot See

The official case for the Pax Silica hub rests on a straightforward arithmetic: investment and jobs on one side, disruption costs on the other, with the balance overwhelmingly positive. That arithmetic is where Section 2's layered history becomes a methodological problem: the communities and ecosystems whose conditions constitute the liability side of this ledger are the same ones the genealogy showed were never fully accounted for under the last arrangement either. The arithmetic is not unreasonable as a starting point. A single AI infrastructure zone capable of attracting sustained capital at the scale the project promises would represent a structural shift in the Philippines' position in the global technology supply chain: a transition from component assembly, where margins are thin and relocation risk is constant, toward the kind of facility-embedded relationship that tends to anchor long-term economic linkage. The employment figures, if realized at anything close to the scale claimed, would be consequential for a provincial labor market of Tarlac's size. Whether they would in fact exceed the contribution of existing regional industries cannot be established from the public record, because no published projection specifies its basis for comparison; the point here is only that the prospective gain is real enough to take seriously, not that its magnitude has been demonstrated. The investment would flow into a host economy that has watched technology-intensive capital bypass it in favor of Vietnam and Malaysia for the better part of two decades. These are genuine prospective benefits. The analysis that follows does not contest them. It contests the method used to weigh them.

The ledger begins with a number, and this is where the analysis must begin. In May 2026, the Philippine News Agency reported over twenty firms planning to invest in the Luzon Economic Corridor (Gonzales, 2026). By June 2026, the Philippine Daily Inquirer was reporting approximately fifty firms interested in the hub (Domingo, 2026). By July 2026, The Philippine Star cited BCDA sources at over thirty firms (Romero, 2026), while a separate BusinessWorld report from the same week placed BCDA's own investment estimate at between \$40 billion and \$70 billion (Sinaking, 2026). Jobs projections have traced a similarly unsettled arc, ranging from 100,000 to more than 190,000 depending on the source and the date.

The span between these figures is not a polling margin; it exceeds a factor of two on the investor count and a factor of almost two on the capital estimate. To average these figures, or to select the most credible, would be to resolve a contradiction that the government itself has not resolved. The range is therefore the finding. When official sources, including BCDA's own communications, cannot agree on the order of magnitude of the investment being hosted, the precision of any downstream cost-benefit calculation rests on an asset side that nobody has pinned down. A ledger that does not know what it contains cannot claim authority on the liability side. The inconsistency does not prove that the benefits are illusory; it proves that the accounting is not rigorously tracked, which is itself a governance condition that the framework agreement must address before the November 2026 signing.

Granting, for the purposes of argument, the most optimistic scenario — fifty firms, \$70 billion, 190,000 jobs — does not resolve the methodological problem that follows. Even a well-sourced and internally consistent ledger can be built on a flawed architecture, and standard cost-benefit analysis as applied to projects of this kind carries a flaw that is not ethical in character but analytical.

The dominant tradition in quantifying environmental and ecological trade-offs is the one that Costanza and colleagues anchored with their 1997 estimate of the aggregate monetary value of the world's ecosystem services: assign prices to what nature provides, add them to the ledger, and the full accounting becomes tractable (Costanza et al., 1997). The appeal of this approach is precisely its apparent completeness: nothing is left outside the frame, every trade-off becomes commensurable, the balance sheet closes. It is the approach implicitly assumed by any project appraisal that offers cash compensation for displacement, monetary offsets for ecological impact, and the assurance that what is lost will be replaced by what is gained. For a range of questions this logic is sound. For others, it commits an error that goes unrecognized as an error because the methodology is applied without examining its foundational assumption.

That assumption is substitutability: the claim that a loss in one domain can be meaningfully replaced by a gain of equivalent value in another. The ecological economics literature has established where this assumption holds and where it breaks down. Daly (1990) identified manufactured and natural capital as complementary rather than substitutable: a factory cannot manufacture clean water from an exhausted aquifer; it depends on

the aquifer to operate. Neumayer (2025) formalized this as the distinction between weak and strong sustainability: weak sustainability permits trading down natural capital provided manufactured capital compensates, while strong sustainability holds that certain natural assets provide services no manufactured substitute can replicate, and must be maintained regardless of the monetary trade-off. Pelenc and Ballet (2015) applied this to the capability approach: when critical natural capital is degraded, what is lost is not merely welfare in the utilitarian sense but the substantive freedoms — food security, livelihood continuity, place-based knowledge — that communities have reason to value and that cash transfers do not restore. Bithas (2020) gave the argument operational content through ecological thresholds: where natural systems have tipping points beyond which degradation is irreversible, the policy question is not how to price the damage after the fact but where the threshold sits and how to avoid crossing it. Applied to the Clark zone, the practical implication is direct: if large-scale industrial water draw pushes the groundwater system serving the site and its surrounding farmland past the rate at which it recharges, no compensation paid to affected communities restores the hydrology. Where that threshold sits for this particular system is not established in any public document, which is precisely why the conditional cannot presently be resolved in either direction. There is only the discovery, after construction, that the cost was never assessed because it was never examined.

The analytical conclusion that follows from this literature can be stated as a taxonomy. Three kinds of costs: money answers the first, engineering the second, nothing answers the third.

The first category is compensable costs. Relocation expenses, lost income during the transition to a new livelihood, the capital required to rebuild a household economy in a different place: these are genuine costs that money can genuinely answer, provided the compensation is calculated honestly against actual losses rather than assessed land values, paid in full and on time, and structured so that it follows the people who bear the costs rather than the properties they formerly held. Assuming those conditions are met, and a framework agreement can be written to require them, the ledger handles this category.

The second category is mitigable costs. Water stress, grid strain, air quality degradation from construction and operation, waste heat at the scale of a large data center cluster: these are also real costs, and engineering design answers them if engineering requirements are written into the project and enforced against verifiable benchmarks. Closed-loop industrial water systems can substantially reduce net water draw on agricultural and municipal sources. Dedicated renewable generation capacity, ring-fenced to zone use, can insulate household electricity supply from industrial demand. These conditions are achievable. A framework agreement that requires them does not block the project; it alters the specifications. The difference between a well-designed AI hub and a poorly designed one is partly technical, and the technical choices are negotiable before ground breaks.

The third category is non-compensable costs, and here the ledger defense fails in a way that is not a matter of calibration. Extinguished ancestral domain claims cannot be compensated because the right secured under IPRA (RA 8371) is not a right to receive money for land; it is a right to the land itself and to the collective self-determination processes that land makes possible. When a domain claim is extinguished by conversion — not resolved through the consultation and consent processes the law requires, but administratively foreclosed — what disappears is not a monetary asset but the legal and material foundation of a form of collective life. No compensation figure restores it, because the claim was never reducible to its market value in the first place. Irreversible ecological loss falls in the same category for the same structural reason: what defines irreversibility is precisely that no substitute can be generated. A watershed degraded beyond its recovery threshold is not replaceable by an ecological fund; it is absent, and the communities who depended on it live permanently with its absence. Foreclosed ways of life — the agricultural practices, intergenerational land relationships, and place-based knowledge of the Aeta Hungey community at Sapang Kawayan and of the farming communities around Capas — are similarly non-substitutable. They can be ended; they cannot be purchased back into existence once ended.

The ledger defense commits an analytical error, not just an ethical one: it prices the unpriceable. Standard cost-benefit analysis converts Category 3 into Category 1 by assigning monetary values to all costs and summing across them. The conversion is silent: the methodology does not mark the point at which it moves from pricing a displaced household's lost income, which is a tractable problem, to pricing an extinguished ancestral domain

claim or the permanent loss of a hydrological function, which are categorically different problems. This is not a data problem that more careful measurement would solve. The assumption of substitutability is structural; it is built into the method before any figures are entered. Applying standard CBA to non-compensable costs does not produce a rough answer improvable with better information; it produces a category error formatted to look like a calculation.

Against this theoretical backdrop, the evidence gap at Clark takes on a significance beyond the procedural. There is no independent hydrological model for the Pampanga basin under the water demand that a large-scale data center cluster would generate. There is no displacement census for the communities in the Capas area who are already facing documented land pressure: the SunStar Pampanga (2026) report on harassment of Capas residents was published in July 2026, against a framework timeline that targets November 2026. There is no published ecological baseline for the 1,618 hectares designated for the hub or for the wider catchment that makes those hectares hydrologically meaningful. And there is no strategic environmental assessment at the scale at which the project's actual environmental footprint operates. On July 20, 2026, Malacañang announced that the Department of Environment and Natural Resources (DENR) is now conducting an environmental impact assessment and will require an Environmental Impact Statement and Environmental Management Plan before implementation (Philippine News Agency, 2026); this is a genuine procedural development, and it should be weighed as such. It is not yet the thing this analysis has been documenting the absence of. An announced requirement for a future study is not a completed, published, independent one, and the analytical question that matters, whether the DENR assessment concludes and is made public with enough lead time to actually inform the November 2026 framework agreement rather than simply follow it, remains open as of this writing.

The hub is one node in a system, and the system is not bilateral. The Luzon Economic Corridor that frames the zone is a formal trilateral framework: the United States, Japan, and the Philippines jointly reaffirmed in a May 15, 2026 statement that the corridor's digital connectivity and advanced manufacturing initiatives "complement the Pax Silica initiative to which our three countries are signatories," and used the same statement to welcome eight further partners, Australia, Canada, Denmark, France, Italy, the Republic of Korea, Sweden, and the United Kingdom, into the corridor's development (U.S. Embassy in the Philippines et al., 2026). Framing New Clark City as a US-Philippines bilateral arrangement understates the negotiating architecture the Philippines actually sits inside; Japan is a co-equal signatory to the corridor framework that Pax Silica's Golden Node depends on for its transport, energy, and digital connectivity, and the corridor's widening membership means the governance terms Manila sets will be read by a larger and more heterogeneous audience than a bilateral framing would suggest. The corridor that frames the zone encompasses induced effects that a project-level environmental impact statement for 1,618 hectares cannot assess by design. Nickel and copper extraction in the region, already operational, will expand in response to the supply chain demand the hub creates; prospective rare earth extraction adds a further layer. Power generation adequate to supply a data center cluster of this scale requires new generating capacity, transmission corridors, and substation infrastructure, each carrying its own land and ecological footprint. Logistics and ancillary industrial activity will follow the zone as they have followed every comparable development. These are not speculative second-order effects. They are the documented pattern of SEZ-adjacent development that Levien (2012) traced in the Indian case and that UNCTAD (2019) identified across the comparative record of approximately 5,400 zones in 147 economies. A project that is one hub in a corridor-scale program cannot be responsibly assessed as if it were a freestanding facility. The cumulative effects at system scale require a system-scale assessment, and no such assessment has been commissioned.

The absence of the studies that would let anyone weigh these costs, at both project and system scale, is itself a governance finding: consent given without them is not informed. This is not a claim that the studies, once conducted, would necessarily show that costs outweigh benefits. They might not. The point is that nobody knows, including the officials approving the timetable and the communities being asked to accept it. Consent given under conditions of systematic ignorance about what is being consented to is not the kind of consent that a framework agreement of this consequence should rest on. The communities of Capas are not being asked to accept a proposal whose costs have been rigorously assessed. They are being asked to accept outcomes whose costs have not been examined at the level of seriousness that a decision of this permanence requires.

The formulation this analysis produces is not complicated. Benefits can be made real; what money can't restore must be protected, not purchased. The jobs and investment the ledger promises are achievable; they are the reason the project exists, and no serious governance argument requires abandoning them. The question is whether the framework governing how those benefits are produced is adequate to the full range of costs it authorizes. A ledger that cannot see non-compensable costs is not a conservative estimate pending future refinement. It is an incomplete instrument being used to authorize decisions that cannot be undone. What the evaluation problem demands, the instrument toolkit must supply: the baseline studies, the FPIC process, the water accounting, the alternatives assessment; each instrument in Section 7 is the direct procedural answer to a gap this section has named.

6

The Framework: Constraints Before Compensation

The evaluation problem identified in Section 5 is architectural rather than arithmetic. Standard cost-benefit analysis is built to compare commensurable quantities, and when it encounters incommensurable ones — ancestral domain, constitutional jurisdiction, the functional integrity of a watershed — it resolves the incompatibility by converting them into commensurable quantities through pricing. That conversion is not announced. It happens at the method's entry point, before any calculation begins, and what emerges carries the structural form of a neutral economic assessment while having already, silently, answered the question the assessment was meant to ask. A framework that simply uses better discount rates or more careful valuations does not solve this problem; it works within it.

The Constraints-Before-Compensation (CBC) framework proposed here addresses the architectural problem by reorganizing the evaluative sequence. Economics is assigned to the domain where it is valid: costs and benefits that are, in principle, commensurable. For the domains where economic analysis is not valid (rights and ecological thresholds), the framework substitutes decision rules matched to the logical character of what is being evaluated. The result is a three-gate structure, applied in strict sequence, in which a project must satisfy each gate before entering the next. Critically, the framework does not import foreign standards into Philippine practice; it sequences obligations that Philippine law already imposes — the consultation and concurrence requirements of RA 7227 § 15, the FPIC mandate of the Indigenous Peoples' Rights Act, and the constitutional provisions of Article XII — into a design discipline that makes compliance visible, sequential, and verifiable. The gates are not weighted criteria in a composite score; they are filters. This sequential, non-bypassable gate structure extends a control pattern the author developed for AI lifecycle governance, where staged Metric, Governance, and Eco gates enforce quantitative, procedural, and sustainability constraints in sequence before a system may proceed to its next deployment stage (Catapang, 2026); applied here, the three gates enforce rights, ecological, and distributional constraints in the same structure. The architecture is rendered as a flowchart in Annex B: Gate 1 leads to Gate 2 leads to Gate 3, with explicit fail-paths that return a project to the conditions that must be met before it can advance.

Gate 1 — Rights Preconditions

The first gate asks whether a project's foundational rights conditions are satisfied. This gate is binary: a project passes or it does not. Nothing within Gate 1 is priced or weighed against projected economic return, because the interests it protects are not available for purchase under Philippine law, or under the moral logic of rights from which that law derives.

Two clusters of conditions constitute Gate 1. The first is free, prior, and informed consent. The binding domestic source of FPIC obligation in the Philippines is the Indigenous Peoples' Rights Act of 1997 (Republic Act No. 8371), which requires genuine consent from affected indigenous communities before development proceeds on ancestral domains. Genuine consent, in the sense IPRA intends, is obtained before the land conversion decision, not secured afterward on a timetable that already assumes conversion will occur. For Pax Silica, this means that for any portion of the 1,618-hectare zone touching ancestral domain applications (including the Aeta Hungey community at Sapang Kawayan), Gate 1 requires confirmation that consent was obtained before site designation was finalized, that ancestral domain claims were resolved prior to land conversion rather than deferred to a remediation process that runs concurrent with construction, and that the consent process was not collapsed into information delivery or consultation on parameters already decided.

The international benchmark for FPIC is ILO Convention No. 169 on Indigenous and Tribal Peoples, which requires prior consultation and consent for measures affecting indigenous lands, resources, and ways of life. The Philippines has not ratified Convention 169; only one state in Asia, Nepal in 2007, has done so, and Philippine ratification remains pending without a firm legislative schedule. That non-ratification does not reduce the IPRA obligation; it confirms that the obligation flows entirely from domestic law. A government that declines international bindingness while its own statute independently requires the same conduct has not gained flexibility; it has made the domestic statute the sole and unambiguous basis of its legal exposure.

The second cluster within Gate 1 is constitutional jurisdiction. Article XII, Section 10 of the 1987 Constitution provides that “in the grant of rights, privileges, and concessions covering the national economy and patrimony, the State shall give preference to qualified Filipinos” and that “the State shall regulate and exercise authority over foreign investments within its national jurisdiction.” The Supreme Court held Section 10, paragraph 2 self-executing in *Manila Prince Hotel v. GSIS*: no implementing legislation is required before its protections attach. Whatever jurisdiction or immunity arrangement the Pax Silica framework agreement contains, its terms must be consistent with Article XII, Section 10. If they are not, the arrangement is unconstitutional irrespective of the economic case for it, and Gate 1 fails regardless of what Gates 2 and 3 might show.

The binary character of Gate 1 is not a rhetorical design choice. It reflects the structure of rights as a category: a right that can be overridden by a sufficiently large economic benefit is not a right in the sense that IPRA or the Constitution uses the term. Converting Gate 1 into a weighted factor, allowing strong economic performance in Gates 2 and 3 to compensate for a weak rights showing at Gate 1, would reproduce exactly the analytical error that makes standard CBA insufficient for this class of project.

Gate 2 — Critical Natural Capital Screening

Projects that pass Gate 1 enter a second gate organized around ecological thresholds: natural assets whose functional integrity, once degraded beyond a tipping point, cannot be restored by any manufactured substitute or financial transfer. The strong-sustainability literature establishes the analytical basis. Pelenc and Ballet (2015) define critical natural capital as assets providing services that cannot be replicated once lost; Bithas (2020) identifies ecological thresholds as the operational indicator for when that point has been reached; Neumayer (2025) demonstrates that the capital-substitution assumption underlying standard cost-benefit analysis breaks down precisely at those thresholds.

Applied to the Clark zone, the assets that meet this definition are the groundwater system underlying the site and the watershed functions that regulate its recharge for the surrounding agricultural communities. The delineation of that system, its recharge rate, and its present extraction load are not set out in any publicly available study identified for this paper, so it cannot be stated here how much margin exists; what can be stated is that the category of asset is the kind Gate 2 is designed to protect, and that assigning it to Gate 3 without the underlying data is a decision made in the absence of the information that would justify it. These are not assets whose loss could be compensated by building alternative water infrastructure elsewhere; they are the baseline conditions on which the viability of both the zone and its surrounding human system depend. Once degraded past the recovery threshold, they are gone in any timeframe relevant to present or near-future generations.

Gate 2 routes these assets through the mitigation hierarchy rather than through valuation. The hierarchy — avoid, then minimize, then restore, then offset, applied in that order — is not invented here; it is the established principle in environmental impact assessment for genuinely non-substitutable resources. The strictness of the sequence matters. Offset is last, not interchangeable with avoid. A project configuration that cannot avoid harm to a critical natural capital asset does not skip to offset and purchase compensatory measures elsewhere; it exhausts minimization first, then restoration, before offset becomes an available option. If no configuration within those three steps can address the harm, the project as proposed fails Gate 2. The project does not then proceed with a compensatory payment attached; it returns to design, meaning a genuinely different siting, footprint, or technical specification, before it can reenter the gate.

This is Daly’s (1990) operational principles of sustainable development applied as a gate condition: harvest rates for renewable resources must not exceed regeneration rates; emission rates must not exceed assimilation capacity; the depletion of non-renewable resources requires the development of renewable substitutes. Applied to Gate 2, a project configuration that imposes draw-down of a critical aquifer beyond its recharge rate fails, not because a value judgment has been made about the relative importance of water and data centers, but because the asset has a threshold that the analysis must respect rather than price.

The critical distinction is what Gate 2 does not do. It does not assign a monetary value to the watershed, multiply it by a discount factor, and compare it against projected revenue from zone operations. That procedure,

which is exactly what a cost-benefit analysis incorporating ecosystem-services valuation performs, is what this gate replaces. Costanza et al. (1997) provided the canonical method for such valuation; Section 5's argument is that the method, however carefully executed, cannot be the right tool for assets whose loss is irreversible, because the substitution assumption the method requires is the assumption the irreversibility defeats. Gate 2 substitutes the mitigation hierarchy for valuation precisely because the hierarchy operates without that assumption.

Gate 3 — Distributional Cost-Benefit Analysis

Only costs and benefits that survive Gates 1 and 2 reach Gate 3. At this stage, the project has confirmed that rights preconditions are met and that its interaction with critical natural capital stays within the mitigation hierarchy. What remains is the domain where economic analysis is both applicable and necessary: compensable and mitigable costs as defined in Section 5's taxonomy.

Gate 3 conducts standard cost-benefit analysis, with one modification that standard practice frequently omits: disaggregation. The relevant questions are not only whether aggregate benefits exceed aggregate costs but who gains, who bears costs, in what amounts, and across what time periods. Benefits are counted only if they are additional rather than displaced from elsewhere in the Philippine economy, and only if the projection methodology does not double-count economic activity as it circulates through supply chains. The wide dispersion across official Pax Silica projections illustrates why additionality discipline matters: approximately twenty firms cited in May 2026, thirty in July 2026, and fifty in June 2026, with employment estimates ranging from 100,000 to over 190,000 depending on source and date (Gonzales, 2026; Romero, 2026; Domingo, 2026; Sinaking, 2026; Bases Conversion and Development Authority, 2026). A ledger that cannot agree with itself on inputs cannot be trusted on outputs.

Disaggregation by group and time period is an analytical requirement, not a political concession. A project that generates large aggregate returns while concentrating displacement costs on one community and distributing employment gains among a different population has not answered the distributional question by netting the two figures. Gate 3 requires that the distribution be visible and attributed before synthesis, so that the deliberative stage that follows can engage the actual trade-offs rather than their aggregate shadow.

Integration — Multi-Criteria Deliberation

A project that clears all three gates reaches a deliberative synthesis. This stage is not a fourth pass/fail gate; it is the process through which residual trade-offs, those that survive the rights, ecological, and distributional filters, are resolved in public rather than through the analytical choices embedded in the prior stages.

The deliberative dimension connects to the structured-deliberation instrument specified in Section 7 (instrument 12), which details the mechanism through which affected communities participate in the synthesis rather than receiving its output. Stakeholder weighting in the synthesis reflects the principle that those who bear project costs have a higher claim on the deliberative process than those who receive its benefits at a remove. What emerges is a decision, recorded publicly, on which trade-offs were made and on what reasoning: proceed, proceed with specified modifications, or return to design. Trade-offs that survive Gates 1 through 3 are made openly, not buried in a discount rate.

The Annex B flowchart renders the complete sequence, with fail-paths at each gate shown explicitly. The fail-paths are not ornamental. A gate with no fail-path is not a gate.

The Objection and Its Answer

The objection that this framework kills all projects deserves a serious answer, because it will come from serious people. The concern is that binary conditions at Gate 1 will block large infrastructure deals in practice, regardless of whether they are theoretically passable, because the political and administrative complexity of genuine FPIC

and ancestral domain resolution before project finalization makes the timeline unworkable for investors who cannot hold capital against an indefinite consent process.

The empirical answer is that the framework does not raise the bar above what Philippine law has already set. A project that has obtained genuine FPIC under IPRA, confirmed constitutional consistency under Article XII, Section 10, and can demonstrate a configuration that stays within the mitigation hierarchy for the zone's critical natural capital, while producing documented additional benefits whose distribution is attributable and auditable, passes all three gates. The framework's contribution is to make that bar explicit, sequential, and unambiguous rather than submerged in a composite assessment whose parameters can be adjusted to produce any result required.

The honest answer is that some projects, in some configurations, will fail. A data center footprint that cannot be sited without degrading aquifer recharge beyond recovery thresholds, on land whose indigenous occupants have not consented, fails Gates 1 and 2. The framework did not kill that project; the project's configuration and the absence of real consent killed it. The framework's contribution was to say so clearly before construction rather than discovering it years later through litigation, displacement conflict, ecological degradation, or political crisis.

The alternative to the CBC framework is not a permissive regime in which projects proceed freely. It is a regime in which projects proceed with unresolved rights conditions, contested legitimacy, and deferred liabilities. The pipeline of future costs does not disappear when a project bypasses its foundational conditions: community litigation, political mobilization, exposure in international human rights forums, legal challenges to the framework agreement's constitutional validity. It accumulates, and it materializes at the worst possible time: when capital has already been committed and exit costs are high. A project that clears all three gates under the CBC framework is a project whose opponents cannot plausibly contest on the grounds the framework addresses. That is the difference between an investment that operates with political durability and one that is perennially vulnerable to the disruption its unresolved conditions invite.

The framework is a legitimacy technology, not a veto.

7

The Instrument Toolkit: Writing Win-Win Into the Agreement

The Constraints-Before-Compensation framework developed in Section 6 establishes which costs require avoidance rather than pricing, and which distributional questions must be answered before any project can credibly claim to produce shared benefit. This section provides the operational layer: twelve instruments, each capable of being drafted into a framework agreement before November 2026, that translate the framework's principles into binding text.

Each instrument below is organized consistently: what it does (mechanism), why Philippine or international law licenses it (basis), and what a clause might say (draft provision sketch). Figures in brackets (percentages, days, radii) are placeholders the parties would negotiate; they appear here to show the structure a provision requires, not to settle its content.

No instrument below imports a foreign innovation at Philippine expense. Several are more accurately described as operationalizing legal obligations the Philippine government already carries under the 1987 Constitution, under RA 7227, or under IPRA. The framing of each instrument reflects that reality: the ask is to comply with existing Philippine law, give it real procedural content, and apply it symmetrically to all parties: investors and communities alike.

1. Phased Development Gates

Mechanism. Later phases of the zone's construction and operation are made legally conditional on published performance against environmental and social benchmarks established at the outset. An independent certifier, named in the agreement, triggers each subsequent phase by publishing a compliance report. No certification, no next phase.

Basis. Phased conditionality is standard in project finance: lenders routinely tie disbursement tranches to verified milestone performance, and the logic applies equally to a host-state's commitment to proceed. If early phases reveal benchmark deviations that are unacceptable in scale, the host retains a contractual basis for pausing rather than inheriting a *fait accompli*. This instrument also addresses the stranded-asset risk identified in Section 4. If the AI infrastructure cycle shortens before later phases are built, phased gates ensure the host is not legally committed to accepting zone expansion on terms negotiated before the macro environment changed. Phasing is, on this framing, insurance for investors as much as protection for communities; it is therefore one instrument where the interests of both sides converge rather than conflict.

Draft provision sketch.

"No construction or operation beyond Phase [n] shall commence until an independent certifier, named in Annex [X], has published a compliance report against the Phase [n] benchmarks set out in Schedule [Y], and that report has been publicly available for not fewer than [60] days."

2. Water Additionality Mandate

Mechanism. Zone operators maintain closed-loop industrial water systems. Net extraction from shared aquifers attributable to zone operations is held to zero against a pre-development baseline. Quarterly water accounting reports are published and publicly accessible.

Basis. Article XII, Section 2 of the 1987 Constitution vests State ownership in all waters of the Philippines and requires that their exploration, development, and utilization remain under full State supervision. A closed-loop requirement operationalizes that ownership by ensuring zone activity does not diminish the public resource. No independent hydrology study for the site was identified in the public record during the preparation of this paper. Whether such a study exists and has not been released, or has not been commissioned at all, cannot be determined from outside the process, and the two possibilities carry different implications: the first is a disclosure problem, the second an assessment problem. Either way the mandate proposed here requires the gap to be closed, and closed publicly, before operations begin. Given that the zone sits in an agricultural region whose farming

communities draw on the same hydrological system, the zero-net-extraction standard is the minimum that makes the project compatible with the constitutional disposition of the resource. This is not a novel standard for large industrial financing: the Equator Principles, the risk-management framework more than 140 financial institutions apply to project finance above a threshold size, already require water-stress assessment and mitigation planning as a condition of due diligence, and the semiconductor sector supplies a direct precedent for what happens when that diligence is absent. Taiwan's Hsinchu Science Park, anchor site for TSMC's fabrication plants, faced production-threatening water restrictions during the island's 2021 drought, prompting emergency trucked-in supply and a public reckoning with how chip manufacturing's water intensity had outpaced regional planning. A closed-loop mandate written into the agreement now is the cheaper version of a lesson the semiconductor industry has already learned once.

Draft provision sketch.

"The zone operator shall maintain closed-loop industrial water systems. Net extraction from shared aquifers attributable to zone operations shall not exceed zero relative to the pre-development baseline. Water accounting reports shall be published quarterly."

3. Energy Additionality Mandate

Mechanism. Energy supply to zone operations is sourced exclusively from generation capacity added after the agreement date and dedicated to the zone. Diversion from the existing grid in a manner that demonstrably raises residential tariff rates in Tarlac Province is prohibited. The scale this instrument must govern is not abstract: BCDA president and CEO Joshua Bingcang stated in a July 23, 2026 Palace briefing that the hub's daily consumption could reach 3 gigawatts at full development (Mangaluz, 2026), a figure that would place a single industrial zone's draw in the range of a small national grid. BCDA has itself pointed toward the additionality principle this instrument would make binding, citing 500 megawatts of committed solar capacity and stating an embedded power plant is planned for the site (Mangaluz, 2026); a stated intention to build renewable-first is not the same as an enforceable additionality requirement written into the agreement, and the difference is precisely what this instrument closes.

Basis. The domestic statutory hook precedes any foreign precedent. Section 2 of RA 9136, the Electric Power Industry Reform Act of 2001, declares it State policy to ensure the affordability of electric power supply, to secure transparent and reasonable electricity prices under full public accountability, and to promote indigenous and renewable energy resources in power generation. Section 73 of the same Act establishes a lifeline rate protecting marginalized end-users, a protection Congress has twice extended. RA 9513, the Renewable Energy Act of 2008, supplies the corresponding supply-side instrument through the Renewable Portfolio Standards, which oblige distribution utilities and retail suppliers to source a mandated percentage from eligible renewable generation, and through priority dispatch for renewable output. An additionality requirement for the zone therefore does not import a foreign standard; it applies the affordability and renewable-sourcing commitments already written into Philippine energy law to a single very large new load. As a practical matter, the additionality requirement also protects the investment case: a zone blamed for household electricity cost increases faces sustained political opposition that is preventable by design. Renewable generation dedicated to the zone, rather than redirected from existing supply, benefits both parties: it demonstrates that the AI hub's energy demand does not come at residents' expense, and it satisfies the Environmental, Social, and Governance (ESG) disclosure standards that institutional investors in the zone's tenant companies increasingly require.

Draft provision sketch.

"Energy supply to zone operations shall be sourced exclusively from generation capacity added after [date of agreement] and dedicated to the zone. No energy shall be diverted from the existing grid in a manner that demonstrably raises residential tariff rates in Tarlac Province."

4. Ancestral Domain Equity Participation

Mechanism. Identified Aeta communities hold an equity stake in the zone development entity, with board representation proportional to that stake. Revenue distributions are paid quarterly in perpetuity, not as a one-time settlement, and cannot be waived by any single representative without community ratification.

Basis. The Indigenous Peoples' Rights Act of 1997 (RA 8371) establishes free, prior and informed consent as the binding domestic standard for decisions affecting indigenous peoples' ancestral domains. Equity participation goes further than FPIC's consent requirement: it converts affected communities from parties to be consulted into ongoing partners with a structural stake in the zone's performance. The Organisation for Economic Co-operation and Development's (OECD) synthesis of leading benefit-sharing practices finds that one-time compensation systematically underperforms equity arrangements over the time horizons that match communities' actual exposure to project impacts (Raderschall, Krawchenko, & Leblanc, 2020). The Yardhura Walani rapid scoping review confirms that treaty benefit arrangements structured as ongoing revenue streams, rather than lump-sum settlements, produce measurably better wellbeing outcomes for indigenous communities over time (Wilkes et al., 2025). Aboriginal title doctrine as analyzed across Canada, Australia, and New Zealand establishes the broader jurisprudential lineage: title is not extinguished by payment but held until consent-based disposition, and ongoing benefit-sharing reflects that continuity (Knafla & Westra, 2010). Equity participation, not compensation: partners, not paid-off bystanders.

Draft provision sketch.

"Identified affected Aeta communities shall hold not less than [X]% of the equity of the zone development entity, with corresponding board representation. Revenue distributions shall be paid quarterly in perpetuity, not as a one-time settlement, and shall not be waivable by any single representative acting without community ratification."

5. Independent Monitoring Body

Mechanism. An Independent Monitoring Body (IMB) is established with a budget appropriated by a party independent of the zone authority, and a mandate to publish all findings. The IMB board includes not fewer than two seats for representatives of directly affected communities and one seat for an academic institution. The body's relationship to the zone authority is strictly at arm's length: the design principle is that no party marks its own homework.

Basis. OECD analysis of environmental compliance systems identifies independence and public reporting as the two factors most consistently associated with effective compliance outcomes; oversight bodies whose budget or mandate flows from the regulated entity systematically underperform against both metrics (OECD, 2009). The United Nations Economic Commission for Europe (UNECE) framework for enterprise environmental monitoring and reporting identifies community participation in oversight as a design element that improves both data quality and public trust in results (UNECE, 2007). Both sources confirm the instrument's central logic: independent oversight is not a symbolic concession but a functional requirement for compliance credibility. A monitoring body that cannot publish without zone-authority approval, or whose budget depends on zone-authority cooperation, is a monitoring body in name only.

Draft provision sketch.

"An Independent Monitoring Body (IMB) shall be established with a budget appropriated by [neutral party] and insulated from zone authority direction. The IMB board shall include not fewer than [2] seats for representatives of directly affected communities and [1] seat for an academic institution. All IMB findings shall be published within [30] days of adoption."

6. Baseline Studies as Condition Precedent

Mechanism. Ground-breaking and any land conversion are contractually conditioned on the public availability of four completed studies: an independent hydrological assessment; a full displacement census of all affected households; an ecological baseline survey; and a strategic environmental assessment covering the induced effects of the zone, including mineral extraction, power generation, and transmission infrastructure within a defined radius. Each study must be independently commissioned and publicly available for not fewer than ninety days before any ground-breaking.

Basis. As Section 5 establishes, the absence of these studies is itself a governance finding: consent given without them is not informed consent. Article XII, Section 2 of the 1987 Constitution requires that natural resource development remain under full State control and supervision; baseline knowledge of what those resources are and how the project affects them is a precondition of meaningful supervision, not an add-on to it. The strategic environmental assessment requirement addresses a gap that a project-level environmental impact statement for 1,618 hectares cannot fill. The zone is one node in a larger system that includes induced mining expansion, power generation buildout, and transmission corridors across the wider Luzon Economic Corridor; only a system-level assessment can evaluate those cumulative effects. This instrument and Instrument 11 form the section's "look before you sign" pair: together they require that options be analyzed and conditions characterized before any physical commitment forecloses them.

Draft provision sketch.

"Ground-breaking and any land conversion shall be conditional on the public availability of: (a) an independent hydrological assessment; (b) a full displacement census of all affected households; (c) an ecological baseline survey; and (d) a strategic environmental assessment covering induced effects of the zone including but not limited to mineral extraction, power generation, and transmission infrastructure within [radius] km. These studies shall be independently commissioned and publicly available not fewer than [90] days before any ground-breaking."

7. Local Content and Technology Transfer Requirements

Mechanism. Zone operators meet escalating Filipino employment quotas, rising from a minimum threshold in the first years to a higher floor by year ten, with a defined target for management-level positions held by Filipino nationals by year seven. A skills-development plan, funded at a defined percentage of annual zone revenues, is required from the outset and reviewed every three years.

Basis. This instrument's basis is constitutional, not aspirational. Article XII, Section 12 of the 1987 Constitution requires the State to promote "the preferential use of Filipino labor, domestic materials and locally produced goods." Article XII, Section 14 requires the State to "encourage appropriate technology and regulate its transfer for the national benefit." The framework agreement's local-content and technology-transfer provisions are therefore not new demands on investors; they are the mechanism by which the Philippine government discharges obligations it already carries under its own foundational law. Framing the negotiation in those terms, as compliance with existing Philippine constitutional requirements rather than as concession extraction, changes its character. It is also a guard against the enclave labor market pattern that comparative SEZ research identifies as a recurring failure mode: zones that generate gross employment figures without producing skills transfer or supply-chain linkages to the host economy (Zeng, 2021).

Draft provision sketch.

"Zone operators shall meet minimum Filipino employment quotas of [X]% in the first [3] years, rising to [Y]% by year [10], with [Z]% of management-level positions held by Filipino nationals by year [7]. Technology transfer obligations shall be documented in a skills-development plan, funded at [%] of annual zone revenues, and reviewed every [3] years."

8. Grievance Mechanism with Teeth

Mechanism. Any resident of a directly affected barangay may file a grievance with the zone authority. The authority must respond within a defined window. Unresolved grievances are referable to a named independent arbitrator whose decisions bind the zone authority. Proceedings are conducted in Filipino and Kapampangan, at no cost to the claimant.

Basis. Framework agreements for investment zones routinely include investor-state dispute resolution: binding arbitration, defined timelines, enforceable awards. The design principle of this instrument is symmetry. If investors get enforceable protections, symmetry demands communities do too. A grievance mechanism that produces non-binding recommendations, published in a language the claimant does not read, on a timeline that outlasts the harm it responds to, is not a remedy; it is a procedural defense against the accusation that no remedy exists. The language and cost-accessibility requirements are not ornamental: a mechanism inaccessible to the people it is meant to serve fails on its own terms regardless of how it reads on paper.

Draft provision sketch.

“Any resident of a directly affected barangay may file a grievance with the zone authority. The authority shall respond within [21] days. Unresolved grievances may be referred to [named independent arbitrator] whose decisions are binding on the zone authority. Arbitration shall be accessible in [Filipino/Kapampangan] and at no cost to the claimant.”

9. Sunset and Review Clauses

Mechanism. The framework agreement is subject to mandatory renegotiation at ten-year intervals. No term is construed as perpetual. Exit provisions specify that upon termination, zone infrastructure becomes Philippine public property at no cost to the State.

Basis. The stranded-asset risk identified in Section 4 operates in both directions. A perpetual agreement signed at one point in the AI infrastructure cycle becomes a liability if that cycle turns; equally, provisions that seemed adequate in 2026 may prove inadequate by 2036 as the zone’s scale and character mature. Mandatory renegotiation windows serve both parties: they allow the host to address provisions that proved insufficient in practice, and they give investors a structured mechanism for adjusting terms to changed conditions rather than resorting to adversarial dispute resolution. Exit provisions that protect the host against inheriting stranded liabilities are standard in infrastructure concession agreements globally; there is no principled reason for a zone framework agreement to be less protective of the public interest on this point.

Draft provision sketch.

“This agreement shall be subject to mandatory renegotiation at [10]-year intervals. No term of this agreement shall be construed as perpetual. Exit provisions shall include [specific host-protection clauses] ensuring that upon termination, zone infrastructure becomes Philippine public property at no cost to the State.”

10. Transparency Floor

Mechanism. The full text of the framework agreement, all environmental monitoring data, and all IMB reports are published by default on a designated government website within thirty days of finalization or publication. Any confidentiality carve-out must be specifically justified in writing and does not cover environmental, social, or financial-performance data.

Basis. RA 7227 Section 15 requires that zone policies be determined upon consultation with the inhabitants of directly affected LGUs. Meaningful consultation requires that the document being consulted on is publicly available. A transparency floor is the minimum precondition that statutory requirement imposes. The logic extends beyond Section 15: in a constitutional order where sovereignty resides in the people, an agreement that

converts 1,618 hectares of public land into an internationally integrated economic security zone should be legible to the people who live adjacent to it. Narrow, justified carve-outs for genuinely sensitive commercial or security information are compatible with that principle; a default of opacity enforced through broad confidentiality provisions is not.

Draft provision sketch.

“The full text of this agreement, all environmental monitoring data, and all IMB reports shall be publicly available on a designated government website within [30] days of finalization or publication. Any carve-out for confidentiality shall be specifically justified in writing and shall not cover environmental, social, or financial-performance data.”

11. Alternatives Assessment as Condition Precedent

Mechanism. Before final design lock-in, the zone authority commissions and publishes an alternatives assessment evaluating: reduced-footprint configurations; relocation of high-impact components, including mineral processing, to brownfield or less sensitive sites; and alternative sites not subject to ancestral domain claims. The published assessment documents the reasons for rejecting each alternative considered.

Basis. The precautionary principle, as operationalized in the strong-sustainability literature, requires that irreversible choices be preceded by genuine evaluation of reversible alternatives. Paired with Instrument 6’s baseline-studies requirement, this instrument constitutes the section’s “look before you sign” pair: Instrument 6 requires that present conditions be accurately characterized before ground breaks; Instrument 11 requires that the full range of possible configurations be genuinely considered before final design forecloses them. The documentation requirement is as important as the assessment itself. An alternatives process that leaves no public record of what was rejected, and on what grounds, cannot be distinguished from a process that considered no alternatives. The published record of rejection reasoning is what gives the instrument its accountability function and what allows subsequent review to determine whether the assessment was genuine or perfunctory.

Draft provision sketch.

“Before final design lock-in, the zone authority shall commission and publish an alternatives assessment evaluating: (a) reduced-footprint configurations; (b) relocation of mineral-processing or high-impact components to brownfield sites; and (c) alternative sites not subject to ancestral domain claims. The published assessment shall state the reasons for rejecting each alternative considered.”

12. Structured Deliberation Before Signing, Backed by a Binding Consent Mechanism

Mechanism. Citizens’ assemblies are convened with affected barangays of Capas and Bamban municipalities and with identified Aeta communities, prior to the framework agreement’s finalization, using a structured deliberative process whose procedural standards are specified in a schedule to the agreement itself. The published record of each assembly documents concerns raised and the government’s written response. The framework agreement cannot be executed without documented LGU concurrence and a published FPIC record for all affected indigenous communities.

Basis. The domestic statutory hook precedes any foreign precedent. RA 7227 Section 15, the provision that authorizes the Clark Special Economic Zone, requires that zone policies “be determined upon consultation with the inhabitants of the local government units directly affected” and that zone creation be “subject to the concurrence by resolution of the local government units directly affected.” H.B. 4663, the filed but as yet unadvanced New Clark City Freeport Area Act, carries an identical LGU-concurrence requirement in its Section 3, which conditions creation of the freeport on “the concurrence of the local government units (LGUs) of the Municipalities of Capas and Bamban, Province of Tarlac.” This instrument gives the existing RA 7227 requirement real procedural content: not a consultation session held after the substantive decisions have been made, but a

deliberative process whose outputs the government is required to respond to in writing before the agreement is finalized. It is, in the first instance, a proposal that the government comply with the statute it operates under.

For the Aeta communities, RA 8371 (IPRA) establishes free, prior and informed consent as the binding domestic standard. The instrument extends that requirement by specifying the deliberative structure that FPIC consultation must take: a process meeting defined standards rather than a meeting whose adequacy is self-certified by the party seeking consent.

The deliberative model finds its most relevant empirical precedents in rural, lower-literacy, land-and-climate-policy contexts that more closely resemble Capas than any European reference case would. Fishkin et al. (2017) document Uganda's first deliberative polls: participants brought together across demographic lines in a context of significant educational diversity, land-pressure controversy, and community fragmentation produced measurably more reflective outcomes than standard consultation formats. Mayega et al. (2021), applying deliberative methods to climate risk and land policy in Eastern Uganda's Bududa and Butalejja districts, found that structured deliberation surfaced community concerns, including sustained anxiety about land loss, that conventional consultation processes had not registered. Lekalake and Buchanan-Clarke (2026), drawing on DemocracyNext's study of citizens' assemblies across Mali, Malawi, The Gambia, and South Africa, identify the design conditions under which deliberative processes build durable legitimacy in African governance contexts, including in settings with limited institutional infrastructure.

The instrument's novel contribution lies in combining the deliberative model's breadth with the binding legal force of FPIC consent doctrine. *Maledu and Others v. Itereleng Bakgatla Mineral Resources* (Constitutional Court of South Africa, 2018) established that communal consent, not mere consultation, is required before a mining right is granted; the court drew a meaningful distinction between a process adequate in form and one adequate in substance. Commentary on the judgment has emphasized that the practical burden it creates is procedural as much as substantive: consultation conducted without a genuine prospect of the community declining is not consultation within the meaning of the statute (Werksmans Attorneys, 2021). Mexico's 2023 Ley Minera reform requires executed consultation agreements as a condition precedent before mining concessions proceed, giving the consent requirement contractual rather than merely procedural force (Decreto, 2023; DLA Piper, 2023). No existing instrument combines sortition-based deliberation with binding FPIC-style consent as a condition for an investment-hosting agreement of this type. This section proposes that combination as the instrument's contribution: a design that shapes the deal rather than auditing it after the fact, and that gives the community participants' conclusions legal weight rather than advisory status.

Draft provision sketch.

"Before the framework agreement is finalized, the Philippine government shall convene citizens' assemblies with affected barangays of Capas and Bamban municipalities and with identified Aeta communities, using a deliberative process meeting the standards set out in Schedule [Z]. The published record of each assembly shall document concerns raised and the government's written response. No framework agreement shall be executed without the documented concurrence of the affected LGUs as required by RA 7227 § 15, and without a published FPIC record for all affected indigenous communities as required by RA 8371."

Cross-cutting insight.

The twelve instruments above differ in their technical specificity, their legal basis, and the actors they primarily protect. What they share is a structural property: each costs far less to write into the agreement now than to impose once construction has begun. A phased gate written into the original agreement costs nothing to negotiate; a phased gate litigated into an existing agreement costs years and strains the relationship it is meant to govern. A transparency floor included from the start is a clause; demanded after disputed data surfaces, it is a confrontation. Baseline studies required as a condition precedent are research; commissioned after harm is alleged, they are forensics. The same logic applies across every instrument. The cost of the protection rises sharply once physical commitment and legal lock-in accumulate.

This is why November 2026 is not a deadline in the ordinary sense. A deadline is a moment by which something must be done before opportunity closes. November 2026 is something different: the moment of maximum Philippine leverage, and it does not return. Before the framework agreement is signed, every provision is negotiable at the cost of a clause. After it is signed, and after ground breaks, renegotiation requires overcoming sunk costs, contractual commitments, and the political difficulty of reopening a deal that both governments have publicly endorsed. Whatever is written into the agreement before signing shapes the terms every subsequent host country is offered. Whatever is left unwritten becomes the absence that the template normalizes.



The Template Effect: Why This Is Bigger Than Tarlac

The Philippine government's position on Pax Silica is not unreasonable, and any analysis that treats it as such misconstrues the structure of the opportunity. The Philippines arrives at this negotiation with an asset that is legally structured, administratively capable through BCDA under RA 7227, and positioned within an existing security architecture that the Supreme Court has upheld. The framework agreement offers something rare for a middle power: entry into an emerging economic bloc at its founding moment, when the terms of membership have not yet hardened into precedent and when the anchor partner genuinely needs the host's participation to succeed. BCDA's own projections, \$40 to \$70 billion in investment and six-figure employment creation, reflect a plausible ceiling if the partnership is designed well (Sinaking, 2026; Bases Conversion and Development Authority, 2026). A framework agreement that attracts investment at that scale, anchored to technology transfer and Filipino labor mandates written directly into the deal rather than assumed to follow from investment, would represent a genuine development achievement. The task is not to dispute that claim but to ask what conditions make it true rather than merely projected.

The template effect is not an argument against Pax Silica. It is an argument about how much the quality of this particular agreement matters, not only for the Philippines but for every country that comes after it.

Pax Silica's architecture is explicitly replicable. The initiative assembled in December 2025 as a direct response to China's rare-earth export controls, designed to establish sovereign AI supply chains across allied and partner territories (Shagina, 2026). New Clark City is the first physical node. It will not be the last. The coalition has expanded rapidly. The State Department's own account of the Second Pax Silica Summit, held in Washington on June 25 and 26, 2026, records ten additional partners signing the declaration — Argentina, Chile, Costa Rica, El Salvador, the European Union, Germany, Greece, Kazakhstan, the Netherlands, and Panama — bringing the total to twenty-four signatories in six months (U.S. Department of State, 2026a, 2026b; see also Brownstein Hyatt Farber Schreck, 2026). The more consequential audience, however, sits outside the bloc. Vietnam, Indonesia, Malaysia, and Thailand appear on none of the State Department's signatory lists, having so far declined or deferred accession while pursuing bilateral minerals arrangements with Washington (U.S. Department of State, 2026a; Asia Society Policy Institute, 2026). Whether their absence reflects deliberate preservation of optionality between blocs or the coalition's outreach not yet having reached them, these are the governments with the strongest reason to read the Philippine agreement closely: they are weighing the same decision Manila has already made, with the benefit of seeing its terms first (Brownstein Hyatt Farber Schreck, 2026; Asia Society Policy Institute, 2026). Whatever Manila negotiates now — the jurisdiction clauses, the environmental review standards, the labor protections, the community consent mechanisms, the revenue-sharing formula — becomes the observable benchmark against which every subsequent negotiation is opened. Whatever Manila signs becomes the floor of every future host's negotiation.

This is not a claim about negotiating strategy or intent on either side. It is a structural observation about how international economic agreements harden. The first signed text in a replicable architecture does not merely record what two parties accepted; it creates the reference document against which subsequent parties must argue for more. Raising the floor requires demonstrating that the previous floor was inadequate, a demand that places each subsequent host in a diplomatically awkward position relative to the bloc anchor. The path of least resistance for each successive negotiation is to accept what the last party accepted. The floor, in practice, tends to hold.

The history of special economic zones illustrates how consequential this dynamic has been across analogous arrangements. The United Nations Conference on Trade and Development's 2019 World Investment Report surveyed approximately 5,400 such zones across 147 economies and found that host-country protections — labor standards, environmental requirements, community rights, revenue-sharing formulas — were genuinely embedded far less often than zone promotional materials implied (UNCTAD, 2019). The gap between what zones promised and what host communities actually received was not primarily a function of bad faith. It was a function of template: first-generation agreements in a given wave set terms that proved difficult to renegotiate zone by zone, country by country, without appearing to threaten the investment proposition itself. Zeng's (2021) World Bank synthesis of what governance design that actually works looks like, and where zone arrangements routinely fail, underscores that successful zones share precisely the governance provisions that zone agreements most commonly omit: independent monitoring, binding environmental standards, community benefit-sharing,

and enforceable review mechanisms. The omissions are not random; they reflect which parties carry the drafting burden at the founding moment and which parties bear the consequences of silence.

Levien's (2012) study of India's special economic zones, published in the *Journal of Peasant Studies*, named the failure mode directly: accumulation by dispossession. Land was converted, communities were displaced, fiscal incentives were extended, and the promised development multiplier failed to materialize in forms that benefited those who bore the costs. The mechanism was not unique to India's political economy, and it was not produced by exceptional circumstances. It was the ordinary operation of zone agreements that contained strong investor protections and weak host protections, agreements that looked, in their legal architecture, very much like the agreements that had been signed before them. Zeng (2015), examining SEZ experience across China and Africa for the World Bank, finds a similar pattern: the divergence between zones that delivered broad development gains and zones that produced enclave economies correlated strongly with governance design choices made before and at the time of founding, not with investment volumes or sectoral characteristics.

Two paths diverge from the same starting point.

The first is the agency demonstration path. A founding host that uses its leverage to demand a high-governance framework — one that embeds phasing gates, water and energy additionality, independent monitoring, FPIC compliance under the IPRA (RA 8371), and sunset and review clauses in the signed text rather than leaving them to future implementing instruments — converts bloc membership into something more consequential than geographic participation. It demonstrates that host-country agency at the founding moment is real and that middle powers can set terms rather than merely accept them. For subsequent hosts, the signed text serves as evidence of what is achievable within the bloc's structure. The governance provisions that determine whether a Golden Node is genuinely transformative or merely extractive are, without exception, provisions that most observers will never read. The agency demonstration path is the one in which those provisions are strong, and the subsequent hosts can point to Manila's text as the standard.

The second is the extraction-template path. Here the AI supply chain supplies fresh terminology for a familiar structure: foreign capital, converted land, communities left with the liabilities, and the host government left holding the residual risk when the investment cycle turns. This is not a prediction, and it requires no imputation of intent. It is what the evidence from the SEZ literature shows happens when agreements are not designed to prevent it. Zeng (2015) documents across multiple regions how often zones have reproduced this outcome, not because anchor investors intended it, but because the agreements were never written to require otherwise. Levien's (2012) framing of dispossession as a mechanism rather than an accident is directly relevant: the mechanism operates through legal architecture, through the provisions that are present and those that are absent, through what the founding text requires and what it leaves to discretion.

The CBC framework and the instrument toolkit are the mechanism by which the agency demonstration path is taken: the United States secures a legally predictable investment environment in a genuine supply-chain node; the Philippines secures enforceable constitutional compliance and real negotiating leverage; the communities most directly affected secure equity, voice, and recourse rather than relocation and a promise. These are not competing interests that the framework must balance at the margin. They are compatible outcomes that a well-designed agreement produces simultaneously, and that a poorly designed one fails to produce for any of them.

The paper does not predict which path is taken. It describes the fork and identifies the conditions under which each is chosen. Those conditions are not opaque. The agency demonstration path requires the November 2026 window to produce a text with the governance depth that only the founding negotiation can write in, before the template hardens. The extraction-template path is what results when the founding text is thin: when monitoring is advisory rather than independent and mandated, when environmental review is project-level rather than strategic, when community consent is consultation rather than a binding precondition, when revenue sharing is promised rather than formulaic. Zeng (2021) observes that zones designed with governance from the outset outperform zones where governance was expected to follow from investment rather than to precede it. The November 2026 window is the moment at which that design choice is made and, once the text is signed, is no longer available.

For Vietnam, Indonesia, and Malaysia, the reading exercise will be direct. They will obtain the signed framework agreement. They will look at the jurisdiction provisions, the environmental review requirements, the

labor standards, and the community consent mechanism. If those provisions are substantive, subsequent hosts can argue for equivalence: they can hold Manila's signed text as evidence that the bloc's anchor partner accepted a high-governance floor, and they can demand the same. If those provisions are weak or absent, subsequent hosts face the harder task of arguing for more than the founding host accepted, not impossible but structurally disadvantaged by the existence of a reference text that recorded something less.

This is why the framework developed in this paper is not Clark-specific. The Constraints-Before-Compensation architecture — rights preconditions before any valuation, critical natural capital screened before any compensation calculation, distributional cost-benefit analysis only for what survives the first two gates — applies to any AI-infrastructure host agreement in the Pax Silica bloc or in analogous arrangements elsewhere. The twelve instruments in the preceding section are mapped to Philippine legal hooks because that is the primary context: RA 7227's existing consultation requirements, the 1987 Constitution's labor and technology-transfer obligations, IPRA's FPIC framework. But each instrument has an equivalent basis in the domestic law of any host that carries environmental standards in its constitutional framework, any host that has ratified UNDRIP or holds equivalent customary-title protections, any host whose zone-enabling statute carries consultation requirements parallel to RA 7227 § 15. The CBC framework is portable because the template-effect problem is general. The governance gap that a well-designed founding agreement closes is not a Philippine gap; it is a structural feature of how zone agreements replicate.

The November 2026 window is bigger than Tarlac for exactly this reason. Whatever governance quality Manila's signed text contains, or fails to contain, travels outward, to the next host, and the host after that, along a replication chain that the founding agreement sets in motion. What that chain carries, durable protections or their absence, is fixed at the founding moment, in provisions that determine what every subsequent host is offered as a starting point.

9

Conclusion: What November 2026 Must Contain

The argument assembled across this paper resolves into three findings that stand together. The first is structural: the Pax Silica framework agreement, as it moves toward finalization, contains large unspecified spaces in jurisdiction, environmental review, water allocation, energy supply, labor standards, and data governance; what fills those spaces will determine whether the deal is what its architects say it is. Sovereignty is not affirmed in a signing ceremony. It accumulates, or erodes, provision by provision: in who adjudicates disputes, in who publishes what data, and in who is in the room when those two questions are settled.

The second finding is analytical. The official defense of the project rests on a compensation logic (jobs, investment, infrastructure) that silently converts costs that money cannot answer into costs that money can. Extinguished ancestral domain claims, irreversible ecological loss, and foreclosed ways of life belong to a category distinct from relocation expenses or grid strain: neither money nor engineering restores them. Treating them as compensable is not merely an ethical position; it is an analytical error. The baseline studies that would make the distinction legible do not exist: no independent hydrology, no displacement census, no strategic environmental assessment covering induced upstream effects. That absence means no one has yet established what is being traded, let alone whether the trade is acceptable.

The third finding is constructive. The Constraints-Before-Compensation framework proposed here addresses both problems simultaneously: rights preconditions before economic calculation; critical natural capital screening before cost-benefit analysis; distributional analysis before signing. The twelve instruments that operationalize it are not additions to the deal; they are the minimum content of a deal that can be defended. Whether the Golden Node becomes an anchor for Philippine development or an enclave with better signage will be settled in the clauses that fewest people read.

Those provisions are what November 2026 must contain.

Minimum provisions, what a legitimate framework agreement includes:

1. **Phased development gates.** No subsequent phase proceeds without published independent certification that environmental and social benchmarks from the prior phase have been met.
2. **Water additionality.** Industrial water systems must be closed-loop; zero net draw on agricultural or domestic sources; published water accounting is a condition of continued operation.
3. **Energy additionality.** New renewable generation dedicated to the zone; no diversion of existing supply in any form that raises household tariffs.
4. **Ancestral domain equity participation.** Aeta communities hold equity stakes with board representation and revenue streams in perpetuity; one-time resettlement compensation does not satisfy this requirement.
5. **Independent monitoring body.** Mandate, budget, and membership structured so the zone authority cannot direct or defund it; community and academic seats filled before operations begin.
6. **Baseline studies before ground-breaking.** Independent hydrology, displacement census, ecological baseline, and a strategic environmental assessment covering induced upstream effects, including mining, power generation, and transmission corridors, all completed and published before construction commences; written into the agreement as conditions precedent.
7. **Local content and technology transfer.** Escalating Filipino employment and management quotas; skills pipeline funded by zone revenues; consistent with 1987 Const. art. XII §§ 12 and 14.
8. **Grievance mechanism with teeth.** Binding remediation timelines and third-party arbitration accessible to affected communities on terms symmetrical with investor dispute resolution.
9. **Sunset and review clauses.** Mandatory renegotiation windows at defined intervals; no perpetual terms; exit provisions that do not strand the host with converted land and idle infrastructure.
10. **Transparency floor.** Framework agreement text, environmental data, and monitoring reports published by default; carve-outs narrow, defined in the agreement, and individually justified.
11. **Alternatives assessment before design lock-in.** Documented analysis of reduced-footprint configurations and brownfield options, with written reasons for rejecting each, completed before final siting decisions are taken.

12. **Structured deliberation before signing.** Citizens' assemblies with affected barangays and Aeta communities, conducted before the framework agreement is finalized; published record of what was raised and how the draft responded; FPIC documentation; LGU concurrence as required by RA 7227 § 15.

A government that signs a framework agreement lacking these provisions does not merely accept a weaker deal for its own population. It establishes, in observed text, the floor from which every subsequent host will negotiate. Vietnam, Indonesia, Malaysia — each observing what Manila accepts — will be handed that floor as the starting point. The Philippine signature will be cited not as precedent for ambition but as evidence of what can be obtained without it.

The project's genuine benefits are not in dispute: the investment interest is documented, the employment projections are real in their scope, the strategic rationale is coherent. Those who would cancel the project treat the costs as if they cannot be constrained; those who would accelerate it treat the costs as if they do not exist. Neither position survives the analysis applied here. The twelve provisions above constrain what is constrainable and protect what cannot be restored. They are not designed to kill the project; they are designed to make the project what its advocates say it already is. The conditional-proceed position places greater demands on the Philippines' government than cancellation does: it requires not only identifying what is at stake but specifying, in enforceable text, the conditions under which proceeding is acceptable.

The Philippines has until November 2026. What is written in the agreement before ground breaks cannot be written afterward. The protections — for its people, its land, and its laws — are either in that text or they are not; what is absent from the signed agreement will not be recovered through monitoring, through renegotiation, or through remediation of losses that could have been prevented. Because whatever the Philippines signs becomes the deal every country after it is offered, what November contains is not only a Philippine question. It is a question about what kind of arrangement this era of AI-infrastructure partnership produces, and whether the countries on the receiving end of the offer had any hand in writing it.

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Title Credits

Cover photograph of the Mount Pinatubo crater lake by Renz Meredor, via [Pexels](#). The title page was designed and formatted by the author.

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Legal instruments

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International

Indigenous and Tribal Peoples Convention, 1989 (No. 169), International Labour Organization. <https://www.ohchr.org/en/instruments-mechanisms/instruments/indigenous-and-tribal-peoples-convention-1989-no-169> The Philippines has not ratified this convention. It has been ratified by 23–24 states, only one in Asia (Nepal, 2007); Philippine ratification has been “under consideration” for years.

United Nations Declaration on the Rights of Indigenous Peoples, G.A. Res. 61/295, U.N. Doc. A/RES/61/295 (Sept. 13, 2007). <https://docs.un.org/en/A/res/61/295> Adopted 13 September 2007; 143–144 in favour, 4 against, 11 abstentions. The United States was one of the four states that voted against, with Australia, Canada, and New Zealand; all four later reversed position. Legally non-binding as a General Assembly declaration.

Comparative (South Africa, Mexico)

Decreto por el que se reforman, adicionan y derogan diversas disposiciones de la Ley Minera, de la Ley de Aguas Nacionales, de la Ley General del Equilibrio Ecológico y la Protección al Ambiente y de la Ley General para la Prevención y Gestión Integral de los Residuos, en materia de concesiones para minería y agua. (2023, May 8).

Diario Oficial de la Federación (Edición Vespertina), Mexico. https://www.diputados.gob.mx/LeyesBiblio/ref/lmin/LMin_ref06_08may23.pdf

Maledu and Others v. Itereleng Bakgatla Mineral Resources (Pty) Limited and Another (CCT265/17) [2018] ZACC 41; 2019 (1) BCLR 53 (CC); 2019 (2) SA 1 (CC) (25 October 2018) (S. Afr.).

Philippine — statutes and constitutional provisions

1987 Const. art. XII (Phil.). <https://www.officialgazette.gov.ph/constitutions/the-1987-constitution-of-the-republic-of-the-philippines/the-1987-constitution-of-the-republic-of-the-philippines-article-xii/> Provisions cited in this paper: - § 2 — State ownership of all lands of the public domain, waters, minerals, and other natural resources; exploration, development and utilization “under the full control and supervision of the State”; agreements permitted with Filipino citizens or 60%-Filipino-owned corporations. - § 10 — Congress may reserve investment areas to Filipinos; “In the grant of rights, privileges, and concessions covering the national economy and patrimony, the State shall give preference to qualified Filipinos”; “The State shall regulate and exercise authority over foreign investments within its national jurisdiction and in accordance with its national goals and priorities.” Held self-executing in *Manila Prince Hotel v. GSIS* (no implementing legislation required for § 10 ¶ 2). - § 12 — “The State shall promote the preferential use of Filipino labor, domestic materials and locally produced goods.” - § 14 — “The State shall encourage appropriate technology and regulate its transfer for the national benefit.”

An Act Amending Republic Act No. 7227, as Amended, Otherwise Known as the Bases Conversion and Development Act of 1992, and for Other Purposes, Rep. Act No. 9400 (2007) (Phil.). — separated Clark Air Base into the Clark Freeport Zone; applies RA 7916 incentive provisions to designated zones.

Bases Conversion and Development Act of 1992, Rep. Act No. 7227 (Phil.). [https://bcda.gov.ph/sites/default/files/documents/RA_7227_0%20\(1\).pdf](https://bcda.gov.ph/sites/default/files/documents/RA_7227_0%20(1).pdf) The controlling statute for New Clark City, not the PEZA law. Created BCDA; § 15 authorizes creation of the Clark Special Economic Zone and requires that zone policies “be determined upon consultation with the inhabitants of the local government units directly affected,” with creation “subject to the concurrence by resolution of the local government units directly affected.”

Indigenous Peoples’ Rights Act of 1997, Rep. Act No. 8371 (Phil.). — the binding domestic source of FPIC obligation in the Philippines; the instrument BAYAN alleges is being bypassed.

Proclamation No. 163, s. 1993 (Phil.) — created the Clark Special Economic Zone and transferred administration to BCDA. (Coverage later expanded by Proclamation No. 803, s. 1996 and Proclamation No. 1035, s. 2006.)

Special Economic Zone Act of 1995, Rep. Act No. 7916 (Phil.). <https://www.peza.gov.ph/special-economic-zone-act> Establishes the general Philippine special economic zone framework; § 46 expressly excludes BCDA and RA 7227 areas from PEZA’s holding, and it is therefore not the governing law for New Clark City.

Philippine — pending legislation (status as of July 26, 2026)

H.B. 9257, 20th Cong. (Phil.). Inserts a new Section 15-D into RA 7227, opening 10% of the total land area of major BCDA-managed zones (expressly including the Clark Special Economic Zone, New Clark City, and Clark Freeport Zone) to development, with 5% allocated to residential and mixed use and 5% to institutional and industrial use, declaring portions alienable and disposable; extends BCDA’s corporate term. Approved by the House of Representatives on second reading May 26, 2026 and on third and final reading June 2, 2026 (271–9–2). No record of Senate concurrence or enactment as of July 26, 2026. Predecessors in the 19th Congress: H.B. 8505, passed on third reading August 2023; S.B. 2647.

New Clark City Freeport Area Act, H.B. 4663, 20th Cong. (Phil.), introduced by Reps. Jaime Eduardo Marc D. Cojuangco and Noel N. Rivera. https://docs.congress.hrep.online/legisdocs/basic_20/HB04663.pdf Would create the New Clark City Freeport Area and a New Clark City Development Authority, applying RA 7916 §§ 30–41. Section 3 conditions creation on “the concurrence of the local government units (LGUs) of

the Municipalities of Capas and Bamban, Province of Tarlac,” mirroring the LGU-concurrence structure of RA 7227 § 15. Filed; no record of committee action or passage located as of July 26, 2026.

Philippine–U.S. agreements

Agreement Between the Government of the United States of America and the Government of the Republic of the Philippines Regarding the Treatment of United States Armed Forces Visiting the Philippines (Visiting Forces Agreement), Phil.-U.S., Feb. 10, 1998 (entered into force June 1, 1999). Allocates criminal jurisdiction primarily to the United States for offences committed by its forces. A reciprocal agreement covering Philippine personnel visiting the U.S. was concluded 9 October 1998.

Agreement Between the Government of the Republic of the Philippines and the Government of the United States of America on Enhanced Defense Cooperation (EDCA), Phil.-U.S., Apr. 28, 2014. — signed in Manila by Defense Secretary Voltaire Gazmin and U.S. Ambassador Philip Goldberg; ten-year term; authorizes rotational access to “Agreed Locations.” Constitutionality upheld in *Saguisag v. Ochoa*, G.R. Nos. 212426 & 212444, 12 January 2016, which held EDCA an executive agreement implementing the Mutual Defense Treaty and the VFA and therefore not requiring separate Senate concurrence.

Mutual Defense Treaty, Phil.-U.S., Aug. 30, 1951. The foundational instrument that the VFA and EDCA implement.

Annex A

Settled vs. Unsettled Provisions Table

Table 1: Settled vs. unsettled provisions in the Pax Silica framework agreement

Provision area	Status	Source	What remains open
Jurisdiction / dispute resolution	UNSETTLED	Chi (2026, May 19); Chi (2026, May 26); Helberg (2026); VFA (Phil.-U.S., 1998/1999); EDCA (Phil.-U.S., 2014); RA 7227 § 15	The VFA already allocates criminal jurisdiction primarily to the US for offences committed by US forces; the May 2026 immunity episode was an attempted extension of that existing arrangement into the civilian-commercial domain, not a wholly unprecedented ask, but one that entered uncharted design space (Chi, 2026, May 19; Helberg, 2026). No published provisions govern dispute resolution between the zone authority and investors, zone employees, or residents. The constitutional challenge precedent (<i>Saguisag v. Ochoa</i> , G.R. Nos. 212426 & 212444, 12 January 2016) upheld EDCA as an executive agreement not requiring Senate concurrence, but did not address civilian-commercial jurisdiction. RA 7227 § 15 establishes a domestic consultation floor but is silent on investor-state and community-state dispute processes.
Land acquisition	PARTIALLY SETTLED / CON-TESTED	BCDA (2026); U.S. Embassy (2026); RA 7227; H.B. 9257 (House 3rd reading June 2, 2026; Senate pending); Beltran (2025); SunStar Pampanga (2026)	BCDA authority under RA 7227 legally constitutes the 1,618-ha zone designation. H.B. 9257 would amend RA 7227 to open 10% of BCDA-managed zones — expressly including New Clark City — to development, declaring portions alienable and disposable. Passage of that measure through the House on June 2, 2026 places it on a timeline that runs alongside the framework agreement’s targeted November 2026 signing; as of July 26, 2026 the sequence in which the two are completed is not yet determined. Aeta Hunguey tribal land claims at Sapang Kawayan (~500 families) and Capas farmer protests indicate contested land status predating the Pax Silica designation (Beltran, 2025; SunStar Pampanga, 2026). A concurrent \$515M stadium project compounds cumulative displacement pressure at the site (Beltran, 2025).

(continued)

Provision area	Status	Source	What remains open
Labor standards	UNSETTLED	1987 Const. art. XII §§ 12, 14	1987 Const. art. XII § 12 (preferential use of Filipino labor) and § 14 (regulation of technology transfer for national benefit) constitute the constitutional floor. No zone-specific labor standards, Filipino employment quotas, or skills-pipeline provisions appear in any published Pax Silica document. Whether these constitutional obligations will be operationalized in the zone agreement is unconfirmed.
Environmental review	UNSETTLED, PROCESS AN-NOUNCED	PHIVOLCS (n.d.-a) Pinatubo hazard map, Pampanga; PHIVOLCS (n.d.-b) Pinatubo hazard map, Zambales; Philippine News Agency (2026, July 20)	No independent hydrological model, displacement census, ecological baseline, or strategic environmental assessment covering induced upstream effects — mining expansion, power buildout, transmission corridors, the wider Luzon Economic Corridor — has been published as of this writing. The site is in the Pinatubo lahar-affected region (PHIVOLCS, n.d.) but no published zone document addresses it. On July 20, 2026, Malacañang announced that DENR is conducting an environmental impact assessment and will require an Environmental Impact Statement and Environmental Management Plan before implementation, with monitoring by the Environmental Management Bureau and Multipartite Monitoring Teams; NWRB separately stated local water sources are sufficient. These are announced procedural commitments, not completed, published, independent studies; whether an EIS is finalized and made public before the November 2026 signing, and whether its findings are allowed to bear on the agreement's terms rather than merely follow them, remains to be seen. A project-level EIS for 1,618 hectares cannot substitute for a system-level cumulative assessment; neither has been published.

(continued)

Provision area	Status	Source	What remains open
Water allocation	UNSETTLED	1987 Const. art. XII § 2	1987 Const. art. XII § 2 places all waters under State authority. No water additionality provisions, closed-loop industrial water requirements, or published water accounting obligations appear in Pax Silica documents. Impact on agricultural and domestic water sources shared with surrounding communities is unaddressed.
Energy supply	UNSETTLED	RA 9136 §§ 2, 73; RA 9513 (Renewable Portfolio Standards)	No dedicated renewable generation requirement, prohibition on diversion of existing grid capacity, or zone-specific energy obligation appears in any Pax Silica document identified in the public record. Philippine energy law supplies applicable general standards (RA 9136 §§ 2, 73; RA 9513), but no zone-specific application of them has been published.
Data governance	UNSETTLED — no framework identified in public record	No provisions governing data sovereignty, data localization, cross-border data flows, AI model governance, or surveillance activity within the zone were identified in the public record. Whether such provisions exist in the draft framework agreement cannot be determined from outside the negotiation.	
Security integration	PARTIALLY SETTLED	EDCA (Phil.-U.S., 2014); VFA (Phil.-U.S., 1998/1999); Mutual Defense Treaty (Phil.-U.S., 1951); Shagina (2026)	EDCA provides a framework for US rotational access to agreed locations; the VFA governs criminal jurisdiction over US forces; the Mutual Defense Treaty is the foundational instrument. These instruments address the military domain. The civilian AI hub raises unanswered questions about the boundary between commercial zone operations and the security architecture, specifically whether civilian AI infrastructure falls inside or outside the agreed-location framework and who determines that boundary.

(continued)

Provision area	Status	Source	What remains open
Revenue sharing	UNSETTLED	BCDA (2026); Sinaking (2026); Romero (2026); RA 7227 § 15	BCDA cites investment potential of up to \$70B (Sinaking, 2026) and reports approximately 30 firms as interested as of July 24, 2026 (Romero, 2026), but investment figures vary significantly across sources and dates (approximately 20 firms, May 18; approximately 50 firms, June 17; approximately 30 firms, July 24), suggesting the ledger is not rigorously tracked. No published formula for revenue sharing between the national government, BCDA, directly affected LGUs, or host communities exists. RA 7227 § 15 requires LGU concurrence for zone creation but specifies no revenue formula.
Monitoring / transparency	UNSETTLED	RA 7227 § 15; OECD (2009); UNECE (2007)	No independent monitoring body has been established. No transparency floor requiring default publication of the framework agreement text, environmental data, or monitoring reports has been confirmed. RA 7227 § 15 provides a statutory basis for consultation and LGU concurrence but does not mandate independent ongoing oversight. Good-practice benchmarks for monitoring mandate design and enterprise reporting are established by OECD (2009) and UNECE (2007); neither appears to have been incorporated into any published Pax Silica instrument.
Ancestral domain / FPIC	CONTESTED	IPRA, RA 8371 (Phil.); Beltran (2025); SunStar Pampanga (2026); ILO Convention No. 169, 1989 (not ratified by Philippines); UNDRIP, G.A. Res. 61/295 (2007)	IPRA (RA 8371) is the binding domestic source of FPIC obligation. No published FPIC compliance record for the Pax Silica zone designation exists. The Aeta Hungey tribe (Sapang Kawayan, ~500 families) faces displacement with no published ancestral domain resolution (Beltran, 2025). The 790th Air Base Group's Joint Training Complex at Crow Valley presents a distinct military land-pressure source adjacent to the AI hub designation (SunStar Pampanga, 2026). The Philippines has not ratified ILO C169; UNDRIP is legally non-binding as a General Assembly declaration.

(continued)

Provision area	Status	Source	What remains open
H.B. 9257 — amending RA 7227 to declare portions of BCDA zones alienable and disposable	PASSED HOUSE THIRD READING, June 2, 2026 (271–9–2); no record of Senate concurrence or enactment as of July 26, 2026	H.B. 9257, 20th Cong. (Phil.)	Inserts a new Section 15-D into RA 7227. Would open 10% of the total land area of major BCDA-managed zones — expressly including the Clark Special Economic Zone, New Clark City, and Clark Freeport Zone, alongside Poro Point, John Hay, and Bataan Technology Park — to development: 5% for residential and mixed use, primarily socialized housing, and 5% for institutional and industrial use, with presidential authority to declare a further 5%. Also extends BCDA’s corporate term. A LEDAC Common Legislative Agenda priority measure; sponsors state it addresses concerns behind an earlier veto and preserves a DENR role in land validation. Legislation converting land to alienable status during the Pax Silica negotiation window is directly load-bearing for the land-tenure and ancestral domain analyses in §§ 2 and 4.
H.B. 4663 — New Clark City Freeport Area Act	FILED, 20th Cong.; no record of committee action or passage located	H.B. 4663, 20th Cong. (Phil.)	Would create a New Clark City Freeport Area and apply RA 7916 §§ 30–41, potentially layering a second governance regime onto RA 7227. H.B. 4663 § 3 carries the same LGU-concurrence structure as RA 7227 § 15. Note: RA 7916 § 46 expressly excludes BCDA and RA 7227 areas from PEZA authority; H.B. 4663’s relationship to that exclusion would require clarification if enacted.

Annex B

CBC Framework Flowchart

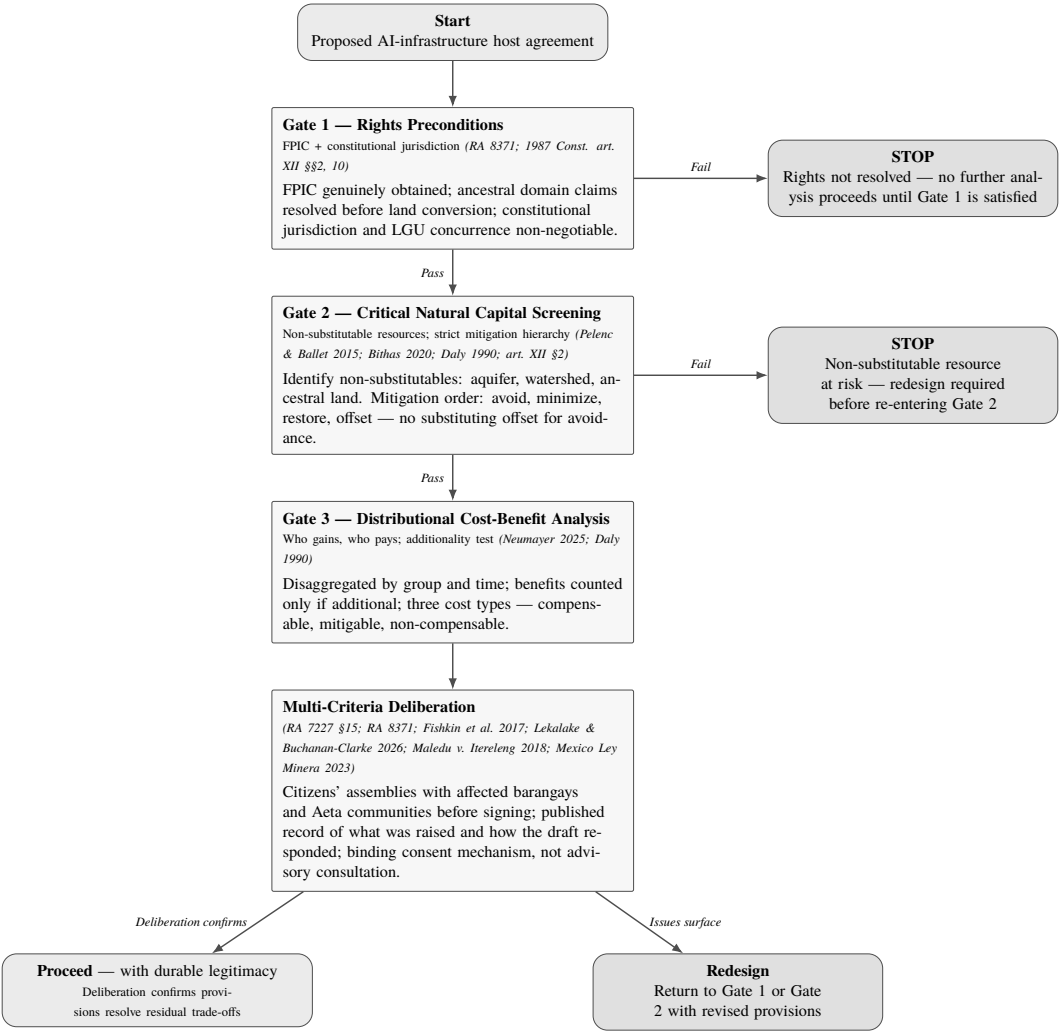


Figure 1: The Constraints-Before-Compensation (CBC) framework: three sequential gates followed by multi-criteria deliberation.

Annex C

Instrument Table

Table 2: The twelve-instrument toolkit: mechanism, legal basis, and draft provision language

#	Instrument	Mechanism	Legal / Precedent Basis	Draft Provision Sketch
1	Phased development gates	Later stages legally conditional on published independent certification of performance against environmental and social benchmarks from earlier stages; certification triggers each phase	Standard project-finance conditionality; RA 7227 § 15 (consultation requirement applies at each stage and should inform phase-release criteria)	”No construction activity beyond Phase [n] shall commence until an independent certifying body, appointed with community and academic representation, has published a compliance report confirming all Phase [n–1] benchmarks have been met and the report has been publicly accessible for a minimum of [30] days.”
2	Water additionality mandate	Closed-loop industrial water systems required throughout zone operations; zero net draw from agricultural or domestic water sources; published quarterly water accounting submitted to and reviewed by the Independent Monitoring Body	1987 Const. art. XII § 2 (State authority over all waters; exploration and utilization under full State control and supervision)	”Net extraction from aquifers and surface water sources serving agricultural or domestic users in directly affected municipalities shall not exceed zero; zone operators shall demonstrate closure of the industrial water cycle through a published quarterly water account submitted to the Independent Monitoring Body within [30] days of the end of each quarter.”
3	Energy additionality mandate	Energy supply sourced exclusively from dedicated new renewable generation capacity constructed after the agreement date; prohibition on supply arrangements that divert existing grid capacity in ways that increase residential or agricultural electricity tariffs	RA 9136 § 2 (affordability, reasonable prices, renewable promotion) and § 73 (lifeline rate); RA 9513 Renewable Portfolio Standards and priority dispatch	”Energy supplied to the zone shall be sourced from generation capacity constructed after the date of entry into force of this agreement; no supply arrangement shall divert generation from the existing grid in a manner that results in increased electricity tariffs for residential or agricultural consumers in directly affected municipalities.”

(continued)

#	Instrument	Mechanism	Legal / Precedent Basis	Draft Provision Sketch
4	Ancestral domain equity participation	Affected Aeta communities hold equity stakes in zone operating entities with commensurate board representation and perpetual revenue streams in place of, not in addition to, one-time resettlement compensation	IPRA, RA 8371 (Phil.); Raderschall et al. (2020) (OECD resource benefit-sharing synthesis); Wilkes et al. (2025) (Yardhura Walani rapid scoping review, treaty benefits and costs); Knafla and Westra (2010) (Aboriginal title doctrine, Canada, Australia, New Zealand)	"Affected Aeta communities, identified through a published FPIC process compliant with IPRA (RA 8371), shall hold not less than [X]% equity in the primary zone operating entity, with board representation proportionate to equity share, and a perpetual revenue-sharing right that survives any sale, restructuring, or transfer of operating rights in the zone."
5	Independent monitoring body	Published mandate and annual work programme; budget appropriated by a neutral party and insulated from zone authority revenues; community, LGU, and academic seats built in by design; all reports published within 30 days of completion	OECD (2009) (environmental compliance monitoring: trends and good practices); UNECE (2007) (environmental monitoring and reporting by enterprises)	"The Independent Monitoring Body (IMB) shall be funded by an appropriation determined by [neutral appropriating party] and insulated from the zone authority's operating budget; its mandate, annual work programme, all monitoring reports, and any dissenting opinions shall be published within 30 days of completion; not fewer than [n] seats on the IMB shall be held by representatives of directly affected barangays, selected through a community-led process."

(continued)

#	Instrument	Mechanism	Legal / Precedent Basis	Draft Provision Sketch
6	Baseline-studies precondition	Independent hydrology, displacement census, ecological baseline, and strategic environmental assessment covering induced upstream effects — all completed and publicly available — are condition precedents to ground-breaking, written into the framework agreement as such	1987 Const. art. XII § 2 (State authority over natural resources and waters); evidence-gap finding (no baseline studies confirmed as published as of drafting: the absence is itself a governance finding)	”Ground-breaking shall be conditional on public availability of: (a) an independent hydrological model covering all shared aquifer and watershed systems; (b) a displacement census covering all directly affected communities; (c) a full ecological baseline; and (d) a strategic environmental assessment covering induced mining expansion, power generation buildout, transmission corridors, and the broader Luzon Economic Corridor. Each study shall be available to the public for a minimum of [60] days before ground-breaking proceeds.”
7	Local content and technology transfer	Escalating Filipino employment and management quotas with separate targets for technical, managerial, and executive roles; mandatory skills pipeline funded by zone revenues; audited annually; safeguards against enclave labor markets that import all skilled roles	1987 Const. art. XII §§ 12, 14 (constitutional obligations to promote preferential use of Filipino labor and to regulate technology transfer for national benefit; this instrument operationalizes existing domestic law, not a new demand); RA 7227 § 15	”Zone operators shall meet minimum Filipino employment quotas of [X]% in Year 1, rising to [Y]% by Year [n], with separate quotas for technical, managerial, and executive roles; a skills-development fund of [n]% of annual zone revenues shall be established, jointly administered by zone operators, affected LGUs, and designated academic institutions, and audited annually by the IMB.”

(continued)

#	Instrument	Mechanism	Legal / Precedent Basis	Draft Provision Sketch
8	Grievance mechanism with teeth	Binding response and remediation timelines; third-party arbitration accessible to community residents on the same procedural terms as investor-state disputes; symmetry is the design principle: if investors hold enforceable dispute rights, communities do too	Equity principle; non-compensable cost finding (Neumayer 2025; Pelenc and Ballet 2015: where costs cannot be priced, procedural access to redress is the operative protection); RA 7227 § 15 (LGU rights in zone governance imply community-level access to dispute processes)	”Any resident of a directly affected barangay may file a formal grievance with the zone grievance officer; grievances shall receive a written initial response within [15] days and a substantive response within [60] days; grievances unresolved within [90] days shall be referred to a third-party arbitration panel on terms not less favorable to the resident than those available to zone investors under investor-state dispute provisions.”
9	Sunset and review clauses	Mandatory public renegotiation windows at defined intervals; no perpetual or indefinitely renewable terms; exit provisions specify published transition plans addressing infrastructure liabilities, employment obligations, and community-revenue continuity	Stranded-asset risk (if the AI capex cycle turns or US industrial policy shifts, no party should be locked into arrangements that cannot be renegotiated; phasing on purely financial grounds justifies this instrument before any ethical argument is made); RA 7227 § 15 (consultation framework implies periodic renewal of community consent)	”This agreement shall be subject to mandatory public renegotiation at [10]-year intervals; renegotiation shall include publication of a performance review and a renewed community consultation process under RA 7227 § 15; exit by either party shall be subject to a published transition plan addressing all host-community liabilities, to be agreed and published no later than [n] months before exit takes effect.”

(continued)

#	Instrument	Mechanism	Legal / Precedent Basis	Draft Provision Sketch
10	Transparency floor	Framework agreement text, all amendments, environmental monitoring data, and IMB reports published by default within 30 days; exemptions are narrow, individually justified in writing, and published in redacted form disclosing the category and date of withheld material	RA 7227 § 15 (informed consultation and meaningful LGU concurrence are impossible without access to the agreement text and monitoring data; the statute creating the zone thus implies default publication by its administrator); democratic accountability	”The full text of this agreement and all amendments shall be publicly accessible within [30] days of execution; all environmental monitoring data and IMB reports shall be published within [30] days of completion; any exemption from full publication shall be individually justified in writing, approved by the IMB, and published in a redacted form that discloses the category and date of the withheld material.”
11	Alternatives assessment	Genuine siting and configuration analysis before design lock-in: reduced-footprint options, brownfield alternatives, relocation of highest-impact components (e.g., mineral processing) to less sensitive sites; documented, published reasons for rejecting each alternative	Precautionary principle; pairs with Instrument 6 as the “look before you sign” pair; RA 7227 § 15 (LGU concurrence on the basis of an alternatives analysis is meaningfully different from concurrence given without one)	”Before final design approval, a public alternatives assessment shall be commissioned and published, covering: (a) reduced-footprint configurations of the zone; (b) alternative sites, including brownfield options; (c) relocation of mineral processing or highest environmental-impact activities to less sensitive sites; with documented, published reasons for rejecting each alternative. LGU concurrence under RA 7227 § 15 shall not be sought until this assessment has been available to the public for a minimum of [60] days.”

(continued)

#	Instrument	Mechanism	Legal / Precedent Basis	Draft Provision Sketch
12	Structured deliberation before signing	Citizens' assemblies convened with affected barangays and Aeta communities before the framework agreement is finalized; published deliberation record forms part of the agreement package; process constitutes a binding consent mechanism — compliance with RA 7227 § 15's LGU-concurrence requirement and IPRA's FPIC obligation — not an advisory consultation	RA 7227 § 15 (zone policies "determined upon consultation with inhabitants of directly affected LGUs"; creation "subject to LGU concurrence by resolution": the instrument gives this existing domestic requirement real procedural content, framed as compliance with Philippine law rather than adoption of a foreign innovation); RA 8371 (FPIC for Aeta communities); Fishkin et al. (2017) (deliberative polling applied in Uganda); Lekalake and Buchanan-Clarke (2026) (citizens' assemblies in Africa: Mali, Malawi, The Gambia, South Africa — rural and lower-literacy contexts closer to Capas than a European referendum); <i>Maledu v. Itereleng</i> (2018) (South African Constitutional Court: communal consent, not mere consultation, required before a mining right); Mexico Ley Minera (2023) (executed consultation agreements required before concessions proceed)	"Citizens' assemblies shall be convened with directly affected barangays and Aeta communities before finalization of this agreement, in compliance with RA 7227 § 15 and RA 8371 as applicable; the published deliberation record shall form part of the agreement package and shall document what was raised, how the draft responded, and what remains unresolved; no final signature shall occur before LGU concurrence by resolution has been obtained as required by RA 7227 § 15 and before FPIC documentation under RA 8371 has been completed and made public."

Annex D

Timeline of Key Events (December 2025 – present)

Table 3: Timeline of key events, December 2025 to present

Date	Event	Source
December 2025	Pax Silica assembled as a direct US response to China's rare earth export controls; the framework draws together allied supply-chain partners and designates anchor zones in partner economies	Shagina (2026)
December 30, 2025	Al Jazeera reports that the Aeta Hungey tribe (Sapang Kawayan, approximately 500 families) faces displacement as the New Clark City "Smart City" development proceeds; a \$515M stadium project is identified as a concurrent, distinct displacement driver adjacent to the AI hub site	Beltran (2025)
April 2026	Philippines accedes to Pax Silica; 1,618-ha (approximately 4,000-acre) designation at New Clark City confirmed; BCDA's role as zone administrator confirmed	U.S. Embassy (2026); BCDA (2026)
April 24, 2026	Daily Tribune reports the Clark AI hub is likely to break ground before 2028	Ayeng (2026)
May 18, 2026	Philippine News Agency reports approximately 20 firms planning to invest in the Luzon Economic Corridor	Gonzales (2026)
May 19, 2026	Philippines rejects a US request for diplomatic immunity at the planned AI hub	Chi (2026, May 19)
May 25, 2026	Under Secretary Helberg disputes the characterization of the immunity episode in a post on X, asserting reporters have "absolutely no idea of what you're talking about" about Pax Silica's purpose	Helberg (2026)
May 26, 2026	Chi reports Helberg's claim that the immunity reporting was "taken out of context"	Chi (2026, May 26)
June 17, 2026	Philippine Daily Inquirer reports approximately 50 firms as keen on the Pax Silica hub	Domingo (2026)
July 15, 2026	SunStar Pampanga reports harassment of Capas residents; the article ties concurrent land pressure to the 790th Air Base Group's Joint Training Complex at Crow Valley, a military footprint adjacent to and legally distinct from the AI hub designation	SunStar Pampanga (2026)
July 24, 2026	BCDA reports approximately 30 firms interested in the Pax Silica project; BCDA pegs total potential investment at up to \$70B. Note: the investment-figure trajectory across sources (approximately 20 firms, May 18; approximately 50 firms, June 17; approximately 30 firms, July 24; BCDA's own \$40–70B estimate, July 24) reflects inconsistent tracking across sources and dates rather than a monotonic trend; the inconsistency is itself an analytical finding.	Romero (2026); Sinaking (2026)

About the Author

Jasper Kyle Catapang is a doctoral candidate in Global Studies at the Graduate School of Global Studies, Tokyo University of Foreign Studies. His technical background spans end-to-end AI development, deployment, and evaluation, and his published research addresses AI ethics, policy, and governance, including a peer-reviewed ethics-by-design control architecture for AI lifecycle governance (Catapang, 2026). This paper applies that dual background, technical fluency in how AI systems are built and evaluated combined with a policy-governance research focus, to the institutional design questions raised by AI infrastructure hosting agreements.

Email: catapang.jasper.kyle.y0@tufs.ac.jp